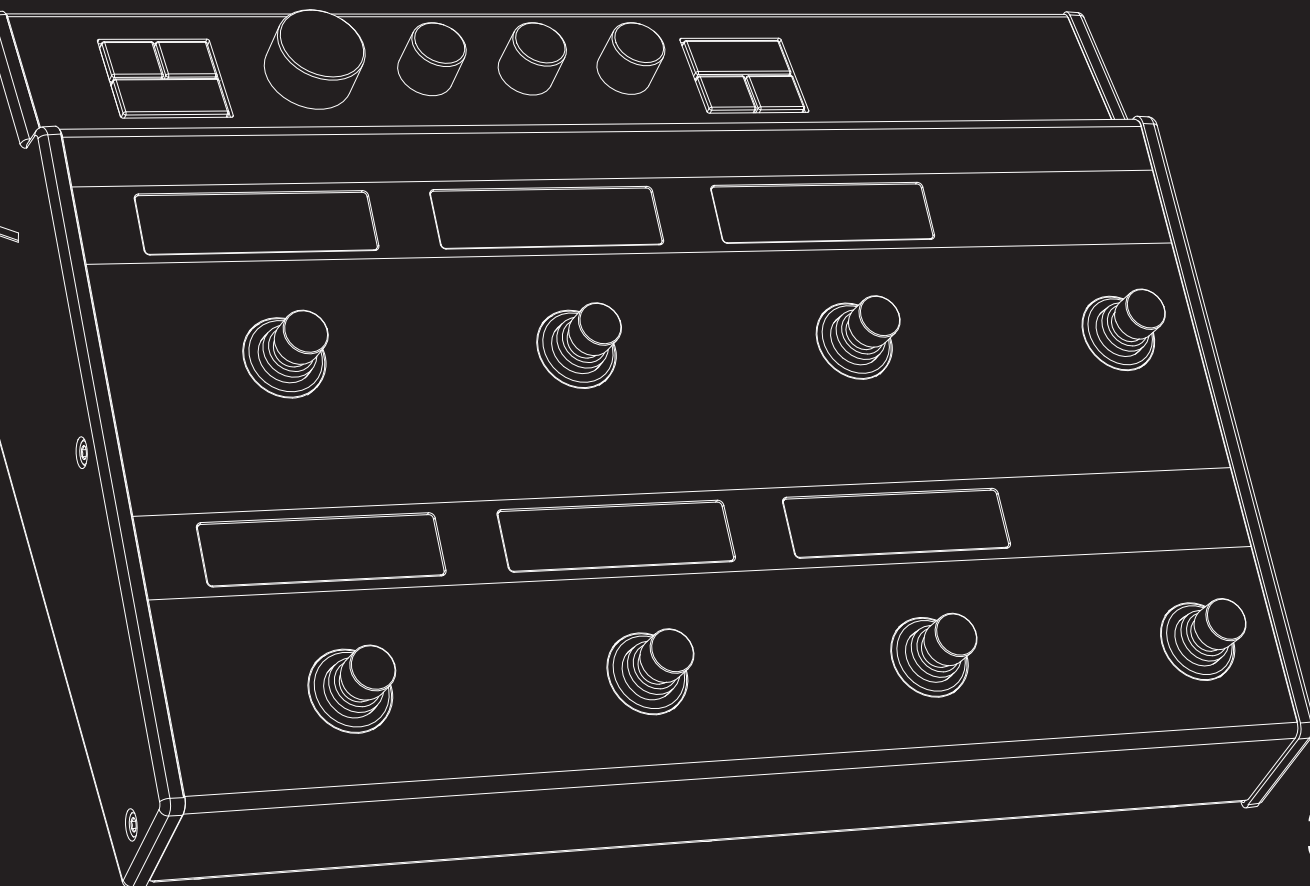




HX EFFECTS



3.80 OWNER'S MANUAL >

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**Supplier's Declaration of Conformity
47 CFR § 2.1077 Compliance Information**

Unique Identifier: Line 6® HX Effects™

Responsible Party - U.S. Contact Information:

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FCC Compliance Statement:

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

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The Model and Serial numbers can be found on the product label. Please record these numbers below and retain this document for your records.

Model Number: _____

Serial Number: _____

Welcome

“I hate reading manuals!”

Us too. We're not overly fond of writing them either, especially when so few people read them. What if we were to make a handful of super-short videos that get you up and running with your Line 6® HX Effects™ device in less than ten minutes, and you promise not to jump online and complain about having to read a manual? Agreed? Go here: line6.com/meet-hx-effects.



“M’eh. Manuals are okay, I suppose.”

All right (deep breath)... Thank you for purchasing the Line 6 HX Effects device, one of the most powerful and flexible audio processors ever created. We hope it helps drive your search for tonal bliss and spawns years of creativity, both on stage and in the studio. This manual covers HX Effects with firmware version 3.80 (and later).

Although the HX Effects unit may appear deep (and it is), it's designed to be used in two ways—one, like any ol' pedalboard where you choose your stompbox, and turn the knobs, and two, like a highly editable and configurable effects workstation.

Although you're likely anxious to rip open the plastic and plug in, wait! At the very least, check out the *HX Effects Cheat Sheet* that came in the box, and keep it handy. Then read the “Quick Start” chapter of this manual, and we'll have you up and shredding in no time. Be sure also to visit line6.com/videos, where we're always adding new video tutorials covering the latest Line 6 gear!

What's In the Box?

- Line 6® HX Effects™ multi-effects device
- *HX Effects Cheat Sheet* (read that one first!)
- AC power adapter
- USB cable
- Warranty card

Common Terminology

While reading this manual, you may encounter several unfamiliar terms. It's important to know what they mean. Be careful—we might toss a pop quiz your way.

Block *Blocks* are objects that represent various elements of your tone, such as individual effects, volume pedals, loopers, signal splits and merges, and Impulse Responses. Each preset can run up to nine simultaneous blocks, DSP permitting.

Model Each processing block can accommodate one model. The HX Effects hardware includes over 220 effects models, from Line 6 Helix® and the DL4™, DM4™, MM4™, FM4™, M13®, M9®, and M5® multieffects devices.

Preset A *preset* is a collection of blocks. It consists of all effects, snapshots, footswitch assignments, controller assignments, and Command Center messages.

Controller *Controllers* are used for adjusting various parameters in real-time. For example, the expression pedal can be used to control a wah, or the mod wheel on your MIDI keyboard can be used to control delay feedback and reverb depth.

Send/ *Sends* and *Returns* are used to insert your favorite stompboxes anywhere in the signal flow or connecting to your guitar amp via “[4-Cable Method](#)”.

Return The HX Effects device has two mono sends and returns, which can be paired for stereo operation.

IR *IRs* (Impulse Responses) are mathematical functions representing the sonic measurements of audio systems. The HX Effects hardware can store up to 128 custom or third-party IRs at a time. See “[Impulse Response \(IR\)](#)”

HX Edit Application



Be sure to grab the free **HX Edit** application for Mac and PC—available for download at line6.com/software.

HX Edit is a full-featured editor and preset & IR librarian which allows you to easily create & restore full HX Effects device backups, as well as update your device to the latest firmware (also see the next section)! You can connect more than one Helix and/or HX device, and HX Edit will display an individual application window for each.

Preset Sharing with Helix/HX Devices & Helix Native Plugin

All Helix and HX hardware and software live within the same ecosystem. Using multiple windows within HX Edit and/or multiple instances of the Helix Native plugin, both blocks and entire presets can be freely dragged and dropped or copied and pasted between units, with some restrictions. Please see the HX Edit and Helix Native [Pilot's Guides](#) for specifics on preset compatibility.

Updating HX Effects Firmware

If your HX Effects device is still running a previous firmware version, it is highly recommended to update to v3.0 (or the latest available version), which can be accomplished easily using the Line 6 HX Edit application!



IMPORTANT! All presets saved and exported from earlier, HX Effects 2.xx firmware versions are fully compatible with firmware version 3.xx. **Please note, however, that presets saved and exported using HX Effects with firmware 3.0 and later will not be compatible with 2.xx firmware versions.** We highly recommend using HX Edit to back up your custom presets **before** updating to firmware 3.xx..

First, install the latest HX Edit software version on your Mac or PC (available at line6.com/software), connect HX Effects to your computer's USB port, and HX Edit will check online and inform you if a newer firmware version is available. If so, HX Edit will then walk you through performing a full device backup and firmware update, all within minutes. Please see the [HX Edit Pilot's Guide](#) for full details.

Marketplace

Be sure to visit the [Line 6 Marketplace online shop](#), where you'll find 3rd-party add-on assets that further enhance the functionality of Helix/HX family devices and Helix Native plugin! As of this writing, 3rd party presets and IRs are available—check back often on Marketplace for product news and announcements.

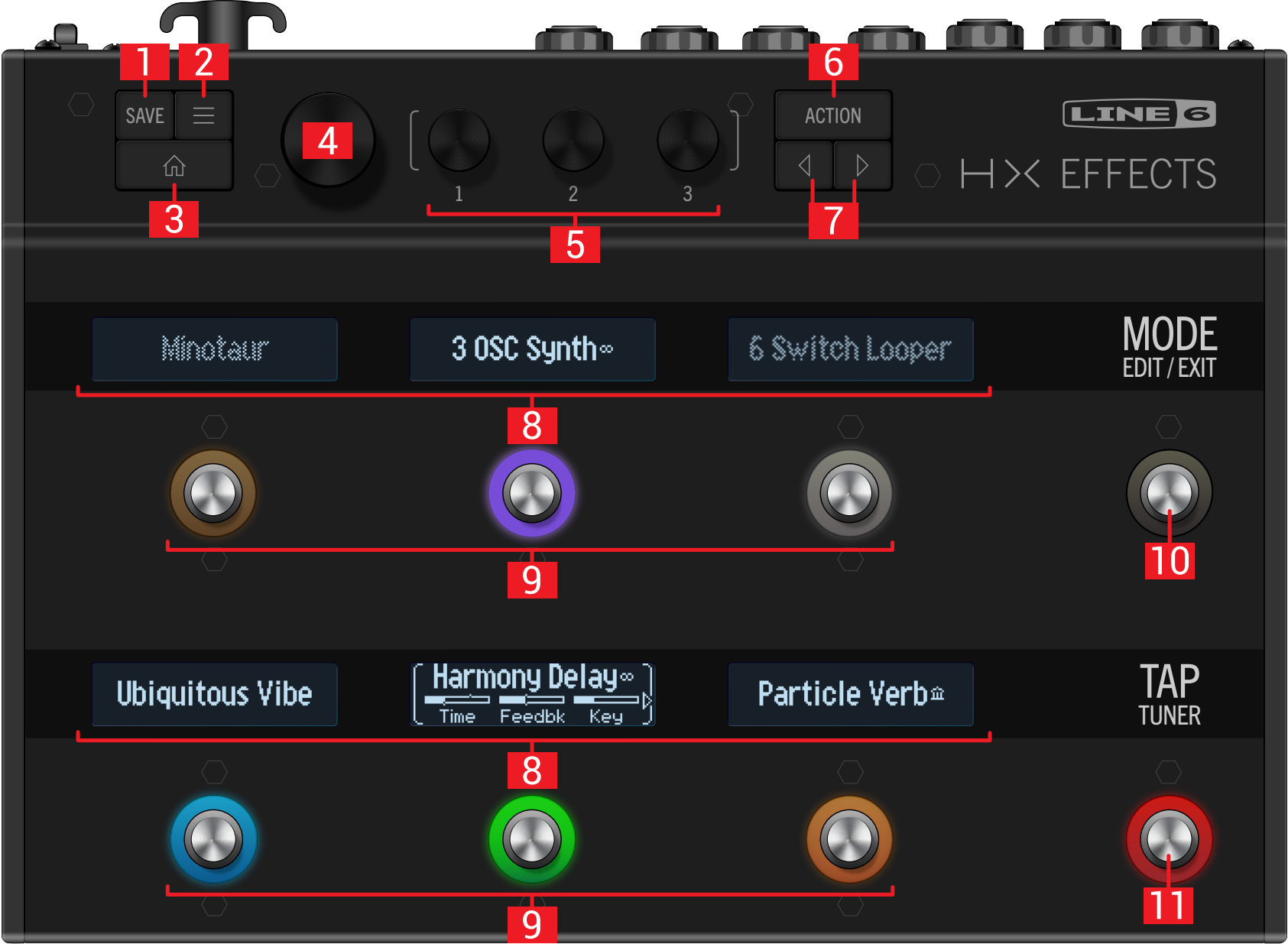


You can access Marketplace directly using the free [HX Edit app](#), where you sign in to your Line 6 online account, then choose to **Get More Presets** or **Get More IRs** from the HX Edit menu. Once you've purchased premium Marketplace assets, they can be downloaded immediately, directly into your HX Edit Presets and/or IR Library, and ready to use on HX Effects! Please see the latest [HX Edit Pilot's Guide](#) for details.



TIP: Registered owners of Line 6 Helix & HX family devices are entitled to a generous discount toward the purchase of the **Helix Native plugin**! Note that all your HX Effects presets and IRs can be loaded directly into Helix Native, making your stage and studio tasks easier than ever! Please visit the [Line 6 Online Shop](#).

Top Panel



1. **SAVE** Press this button to open the Save menu for renaming and saving changes to a preset. Press twice to quick save. See [“Saving/Naming a Preset”](#)
2.  Press this button to dive deeper into the HX Effects device, and you'll find Signal Flow, Controller Assign, Command Center, and Global Settings menus.
3.  If you ever get lost, press this button to return home.
4. **BIG KNOB** From Stomp footswitch mode, turn this knob to change the selected block's effect. Press the Big Knob to open the model list. See [“Choosing an Effect”](#) From Preset footswitch mode, turn the Big Knob to select presets. See [“Preset Mode”](#)
5. **Knobs 1-3** Turn one of the three smaller knobs to adjust parameters for the selected LCD (the one flanked in white brackets); press the knob to reset the parameter's value.



SHORTCUT: For most time-based parameters such as delay time or modulation speed, press the knob to toggle between setting the value in ms or Hz and note divisions (1/4-note, dotted 1/8-note, etc.).



SHORTCUT: Controllers can be assigned to most parameters. Press and hold a parameter's knob to quickly jump to the [“Controller Assign”](#) page for that parameter.

6. **ACTION** Press this button to open the action panel for the selected block or menu. From Stomp footswitch mode, the action panel lets you move, copy, paste, and clear blocks, customize a footswitch's label and LED color, as well as add any changes you've made to a block as a Favorite or save it as a model default. Other menus may have unique action panels. For example, the Global Settings action panel lets you reset all global settings at once.
7.   If the selected model or item has more than one page of parameters, arrows appear in the lower left and/or right corner of its LCD. Press  or  to view more parameters. In the model list, press  to open the folder or  to close the folder and go back one level.
8. **Scribble Strips** Each of the six LCD scribble strips displays the name of its assigned effect or item, so there are never any surprises during a show. You can custom label scribble strips. See [“Customizing a Footswitch”](#). (And don't forget to remove the protective film covering these strips on your new HX Effects device!)



NOTE: If a particular footswitch has more than one block or item assigned, the scribble strip may read “MULTIPLE (X),” where X is the number of assignments. See [“Assigning Switches in Signal Flow View”](#)

9. **Footswitches 1-6** These capacitive, touch-sensitive footswitches have colored LED rings that indicate the effect or item's current state. While in Stomp mode, briefly touch (but don't press) a footswitch to quickly select the assigned effect. Touch the switch repeatedly to cycle through multiple assigned items. See [“Stomp Mode”](#)



SHORTCUT: While in Stomp footswitch mode, touch (but don't press) a footswitch for one second to spill all of its parameters across multiple scribble strips.



SHORTCUT: While in Stomp footswitch mode, touch and hold (but don't press) two switches to swap all assignments between them (including any custom scribble strip names and LED colors). Note that touch-to-select can optionally be turned off, and other footswitch settings customized—see [“Global Settings > Switches”](#) on page 50.

10. **MODE/EDIT/EXIT** Switch Press MODE to toggle between Stomp and Preset footswitch modes. Hold MODE for one second to engage Pedal Edit mode, which lets you edit effects with your feet, allowing your hands can stay on the guitar where they belong. See [“Hands-Free Editing \(Pedal Edit\)”](#) Press MODE/EDIT/EXIT to exit Looper or Pedal Edit mode. See [“Looper”](#)
11. **TAP/TUNER** Switch Press TAP two or more times to set the BPM (beats per minute) of any tempo-based effects such as delay or modulation. Press TAP once to restart any LFO-based modulation effects. Hold TAP for one second to open the Tuner view. See [“Tuner”](#)



SHORTCUT: Touch (but don't press) TAP to display the tempo panel briefly. This lets you quickly fine-tune the current tempo without navigating to the Global Settings menu.

Rear Panel



12. PEDAL/EXT AMP 1-2 Two expression pedals can be connected to the HX Effects device and assigned to adjust a wide variety of parameters. When adding blocks within the Signal Flow view, EXP 1 is automatically assigned to control and bypass Wah, Pitch Wham, or Poly Wham blocks, and EXP 2 is automatically assigned to control Volume and Pan blocks, making them ready for use with your connected expression pedal(s). See ["Assigning a Controller"](#)

Alternatively, either jack can be connected to your traditional guitar amp to switch its channels or turn its reverb on and off. Use a TRS cable for dual operation (A=tip, B=ring). To set this jack's function. See ["Global Settings > Preferences"](#)

13. SENDS/RETURNS 1-2 These 1/4" ins and outs can be used as FX loops for inserting external stompboxes between specific blocks in the HX Effects device or as inputs and outputs for running 4-Cable Method rigs. See ["4-Cable Method"](#)

14. INPUT L/MONO, RIGHT Connect your guitar, bass guitar, or mono pedals to the L/MONO input. Connect stereo pedals, keyboards, synths, or modelers to both the L/MONO and RIGHT inputs.

15. OUTPUT L/MONO, RIGHT Use unbalanced 1/4" TS cables to connect to your guitar amp or other pedals. When connecting to a mono pedal or single amp, connect only the L/MONO 1/4" jack.

16. MIDI IN, OUT/THRU Connect the HX Effects hardware to your MIDI gear to send and receive program changes, continuous controllers, and other MIDI messages. (Note that MIDI communication is optionally also sent and/or received via USB.)

17. USB Connect the HX Effects device to your Mac or Windows computer for preset management and IR loading via the HX Edit application, updating to the latest firmware, receiving MIDI control messages, and sending remote Command Center messages to control various software. Use of a USB 2.0 or 3.0 port is required - Do not use an external USB hub.

18. DC In We recommend using only the included power supply. The provided adapter serves as the disconnect device.

19. Cable Tie Wind the power supply cable through one or both hooks to keep drunk punters from stopping the show.

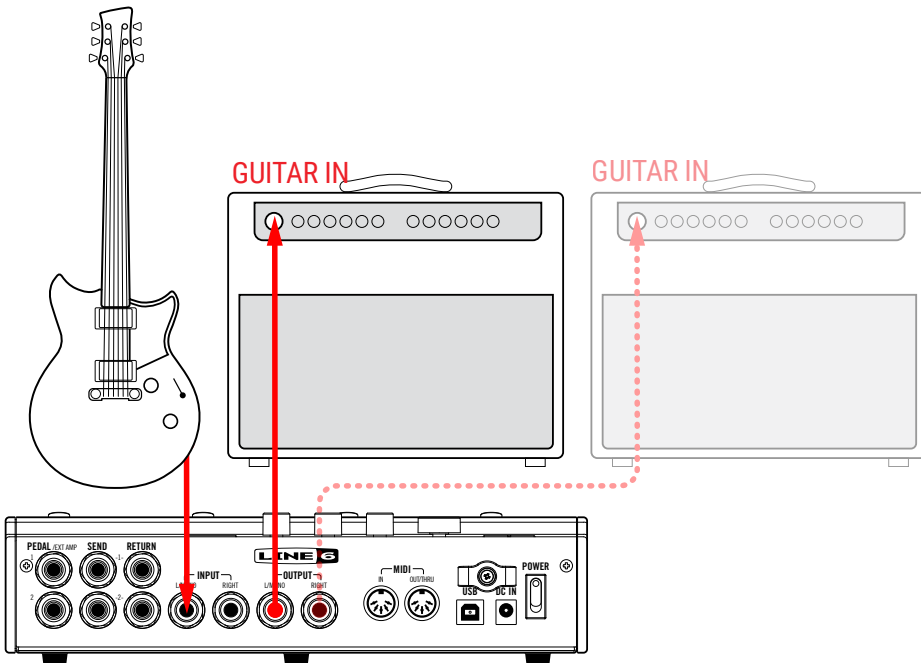
20. POWER switch It's alive!

Quick Start

Hooking It All Up

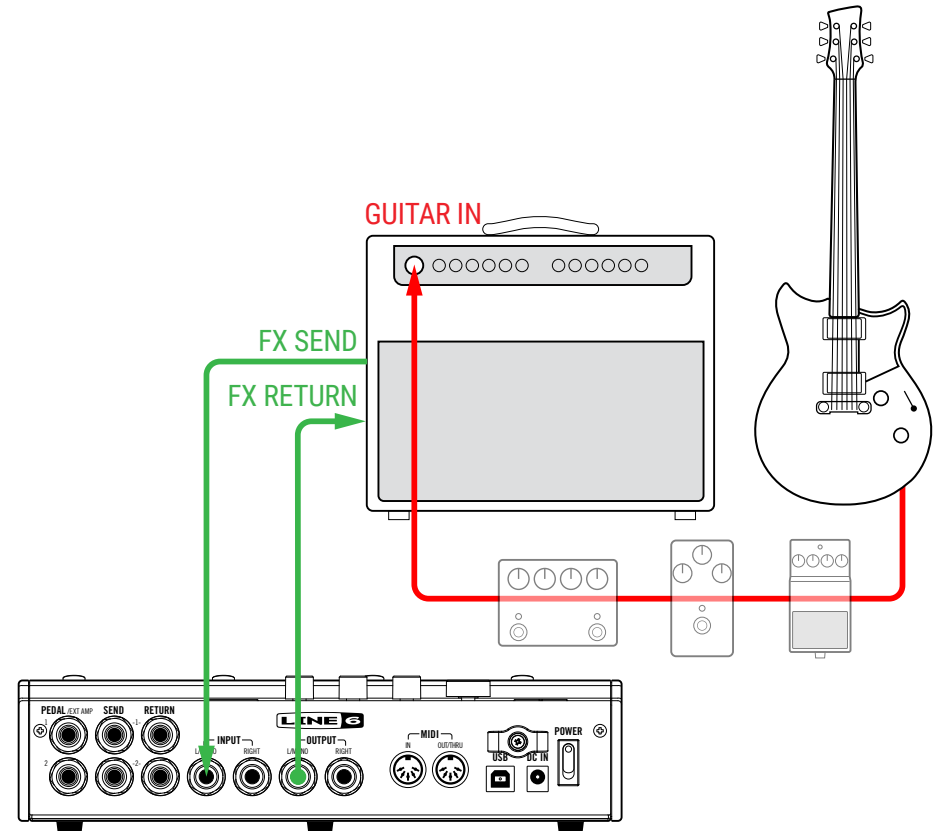
HX Effects as a Traditional Pedalboard (Pre Effects)

Most people will use the HX Effects unit as a traditional effects “pedalboard,” straight into their amp. The HX Effects device can optionally feed a second amp for stereo operation.



HX Effects in your Amp's Effects Loop (Post Effects)

Some guitarists prefer to place time-based effects such as delay and reverb (sometimes called “post” effects) between the preamp and power amp sections of their amp. If your amp has an effects loop, the HX Effects device can accommodate these setups with ease, with or without additional pedals (such as overdrive, fuzz, and compression) before the amp.



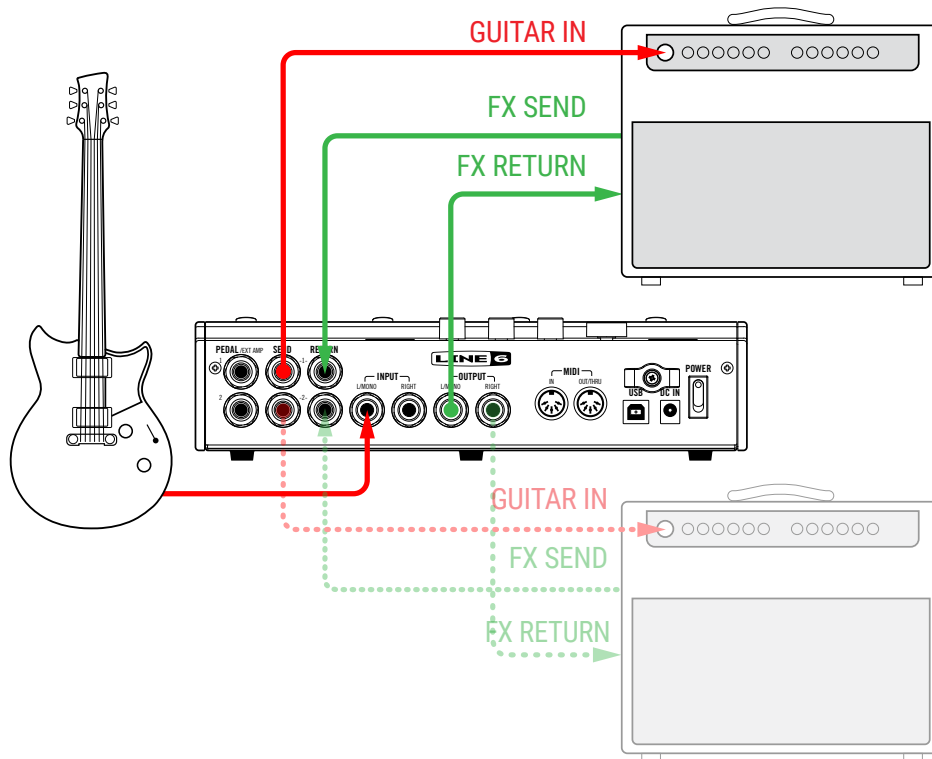
NOTE: The effects loops on most amplifiers run at instrument level. If your amp's effects loop happens to run at line level, you should set the inputs and outputs of your HX Effects device to line level as well. See [“Setting Proper Levels”](#)

4-Cable Method

Of course, the most flexible way to hook things up is with some effects (typically drives, wahs, and compressors) before your amp's preamp and others (the aforementioned delays and reverbs) in its effects loop. This is commonly called 4-Cable Method, and fortunately, your HX Effects device can do this too.

Look for the included factory presets titled with "4CM" to use as your starting template, or you can build your own by adding an FX Loop block (see ["Choosing an Effect"](#)).

Oh, and if you have two amps, your HX Effects hardware can even do 7-Cable Method in stereo.



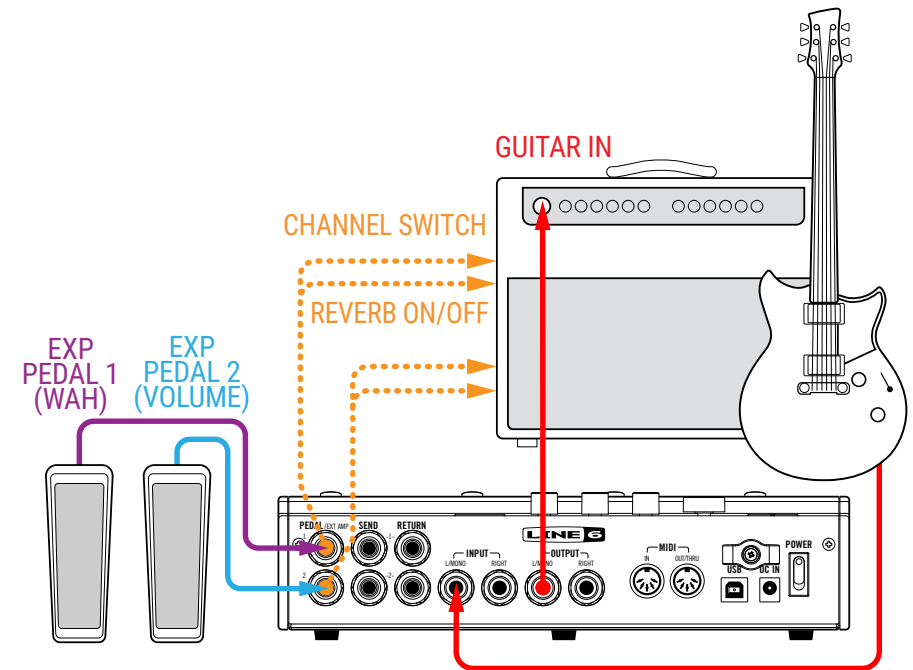
Expression Pedals and/or External Amp Control

By default, the two PEDAL jacks are set up to accommodate an expression pedal each, with PEDAL 1 automatically assigned to any Wah, Pitch Wham, or Poly Wham block and PEDAL 2 automatically assigned to any Volume or Pan blocks. Alternatively, you can choose one or both PEDAL jacks to act as dual (A=Tip, B=Ring) amp control outputs for changing the channels of your amp or turning its reverb on and off.*



TIP: When the blocks are added within the Signal Flow view, PEDAL 1 is automatically assigned to any Wah, Pitch Wham, or Poly Wham blocks, and PEDAL 2 automatically assigned to any Volume or Pan blocks—also see ["Assigning a Controller"](#)

To determine whether each PEDAL jack functions as an expression pedal input or dual amp control output, see ["Global Settings > Preferences"](#)



***IMPORTANT!** Connect EXT AMP 1/2 only to amplifiers that utilize "short-to-sleeve" foot-switch inputs. Connecting to any other sort of input could cause permanent damage to both your amp and HX Effects hardware! If you're not sure if your amp has short-to-sleeve inputs, contact the manufacturer.



IMPORTANT! Your ability to control external amp channel and/or reverb switching with the HX Effects device has been tested with many popular amps and heads. Unfortunately, this does not guarantee compatibility with all products. Note that, depending on the circuitry of the channel switching jack in the guitar amp used, the EXT Amp function may not operate as expected.

Stomp Mode

Stomp mode is where you'll spend most of your time, as it most closely resembles a traditional pedalboard's feel and behavior.

While in Stomp mode, footswitches 1-6 can each do a number of things:

- Toggle one or more effects on and off
- Toggle between two values of one or more parameters
- Generate a MIDI or QWERTY hotkey message, or switch channels on an external amp
- All of the above, even simultaneously

If not already there, press **MODE** to select Stomp mode.

Footswitches 1-6 display any assigned effects, parameter names, Command Center messages, and/or custom labels:



TIPS: Use the Global Settings > Footswitches > Stomp Mode option to configure either all six of the left switches as stomp switches ("6 Swtch"), versus the left two switches as Bank/Preset/Snapshot Up-Down and 4 stomp switches (4 Swtch").

You can also utilize the Command Center - HX Preset, Snapshot, and Looper commands to customize your Stomp mode switches for even greater control of device functions. See ["HX Preset, Snapshot, and Looper Commands" on page 45](#).

Bypassing an Effect

Press the assigned switch.

Bypassed effects display grayed out text and a dim LED ring:



NOTE: Although most people will assign one effect per footswitch, the HX Effects hardware lets you turn multiple effects on and off simultaneously—or even toggle between two or more effects—with a single footswitch. Assigning more than one effect to the same footswitch is accomplished from the Signal Flow menu. See ["Assigning Switches in Signal Flow View"](#)

Bypassing All Effects

1. Press **MODE** and **TAP** simultaneously.

While the HX Effects device is completely bypassed, a warning dialog slowly jumps across all six scribble strips:



NOTE: There are two types of All Bypass for the HX Effects device: Analog bypass (sometimes called "true bypass"), where mechanically switching relays route your signal directly from the inputs to the outputs with no processing or A/D/A conversion, and DSP bypass, where any delay echoes and reverb tails decay naturally. By default, the HX Effects hardware is set for Analog bypass, but this behavior can be set from ["Global Settings > Preferences"](#)

2. Press **MODE** and **TAP** simultaneously again.

The HX Effects device returns to normal operation.

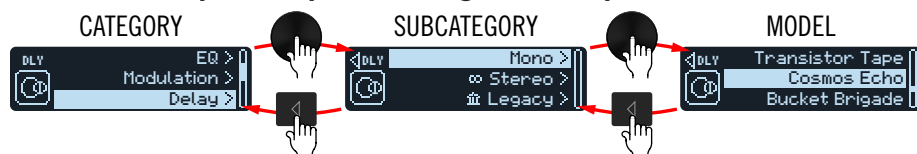
Choosing an Effect

To change an existing effect, briefly touch its footswitch and then turn the Big Knob.

To add an effect to an empty footswitch, briefly touch the switch and then turn the Big Knob.

Selecting effects within the same category—say, changing a Simple Delay to a Reverse Delay—can be very quick. However, as the HX Effects device has almost 200 models to choose from, using this method to change a mono distortion (beginning of the list) into a stereo looper (end of the list) is very slow. Instead, you should open the model list:

1. From Stomp mode, press the Big Knob to open the model list:

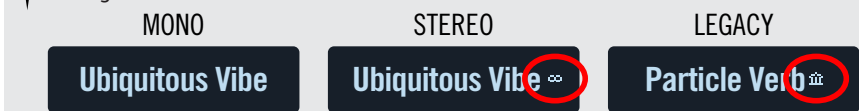


NOTE: Once you've added one or more blocks as "Favorites," you'll see a Favorites option at the top of the Category menu—see ["Adding a Block to Favorites"](#) on page 14.

SHORTCUT: Optionally, you can quickly skip through model subcategories without ever having to open the model list—With the desired block selected within Stomp mode, press and turn the Big Knob to skip subcategories, then turn normally to select models within that subcategory.

The Effects categories include one or more subcategories. Effects in the "Mono" and "Stereo" subcategories are HX (Helix) models. Effects in the "Legacy" subcategory are taken from classic Line 6 pedals, including M13, M9, M5, DL4, MM4, FM4, and DM4.

NOTE: Stomp switches with stereo or legacy models display small identifier icons to the right of the model name:



Turn the Big Knob to select items in a list.

Press the Big Knob (or $\triangleright$) to view a category or subcategory's contents.

Press $\triangleleft$ to go back one level.

IMPORTANT! If certain items don't appear within the model list, this means the HX Effects device cannot accommodate that category, subcategory, or model. For example, if you've already added three reverbs and a pitchshifter, you may not be able to add another pitchshifter—see ["Dynamic DSP"](#)

2. Using the Big Knob, $\triangleleft$, and $\triangleright$, select the desired category, subcategory, and model.
3. To close the model list, press HOME .

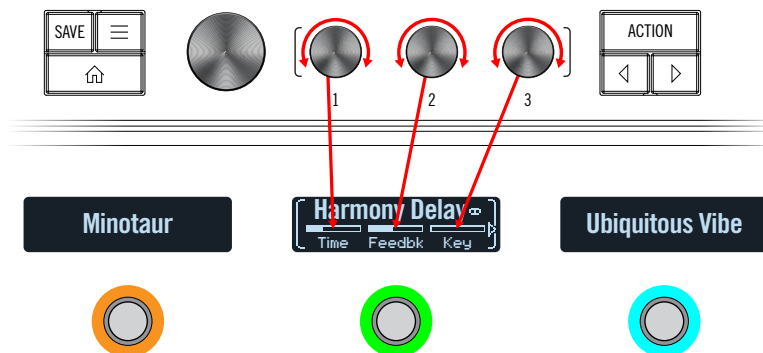
Editing Effects

1. To select an effect for editing, touch its footswitch briefly.

The first three parameters appear and the scribble strip displays white brackets, indicating its knobs control the parameters:



2. Turn Knobs 1-3.



While adjusting parameters, their values briefly appear on the scribble strip:

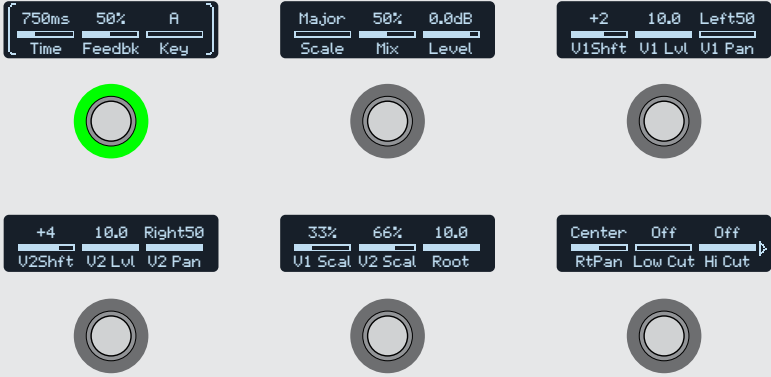


If an effect or other item has more than three parameters, arrows appear to the right and/or left of the value bars (see above).

SHORTCUT: For most time-based parameters such as delay time or modulation speed, press the knob to toggle between setting the value in ms or Hz and note divisions (1/4-note, dotted 1/8-note, etc.).

3. Press ◀ or ▶ to access more parameters (if available).

 **TIP:** Touch a stomp switch for one second to spill all parameters across multiple scribble strip LCDs to view them more easily:



Touch the desired footswitch to select its parameters and then turn knobs 1-3 to adjust them. If a block has more than 18 parameters, press ▶ to access more parameters. When finished, touch a switch for one second (or press ⏮) to exit.

Hands-Free Editing (Pedal Edit)

Effects can be edited without taking your hands off the guitar. If you hate having to reach down and twist knobs, Pedal Edit mode will be your new best friend. Although Pedal Edit is not intended to replace proper controller assignment, in a pinch, it can also be used to tweak one parameter at a time during a performance.

1. Hold MODE (EDIT) for two seconds.

The preset's processing blocks appear on the footswitches, flashing. If the preset has more than six blocks, FS3 displays "MORE..." Press FS3 (MORE...) to view more blocks.

 **NOTE:** Selecting a block in Pedal Edit mode may not correspond to any footswitch assignment it may have.

2. Press the switch of the effect you want to edit.

The block's first page of parameters appear on footswitches 1-3:



If the effect has more than one page of parameters, press FS4 (PAGE >) until you find the parameter. To go back one page, press and hold FS4.

3. Press the footswitch displaying the parameter you want to adjust.

Press and hold a Time or Speed switch to toggle between setting the value in ms or Hz and note divisions (1/4-note, dotted 1/8-note, etc.).

Press and hold any other parameter switch to reset its value to default.

4. Use a connected expression pedal to adjust the parameter's value.

For fine adjustment, press FS5 (VALUE-) and FS6 (VALUE+). Hold FS5 (VALUE-) or FS6 (VALUE+) for faster adjustments.

5. When finished, press MODE (EXIT).

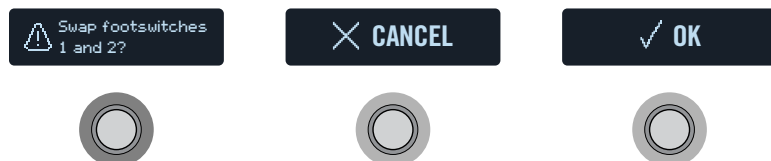
If you want to save any changes made to the preset, hold MODE (EXIT) for two seconds.

Swapping Footswitches

If you want to change the location of Stomp mode footswitches, you can quickly swap two footswitches.

IMPORTANT! Swapping footswitch locations does not affect the actual order of effects. To reorder your effects, see ["Moving Blocks in Signal Flow View"](#)

1. Touch any two Stomp mode footswitches until the following options appear:



2. Press the OK switch.

Copying/Pasting an Effect

Effects can be copied and then pasted onto another switch either in the same preset or an entirely different preset.

1. Touch the stomp switch you want to copy and press ACTION.
Action switches appear around the selected effect.
2. Press the COPY BLOCK switch.
3. Touch a switch you wish to paste the block onto—even in a different preset—and press ACTION.
4. Press the PASTE BLOCK switch.

NOTE: There may be situations where a block cannot be pasted into the selected location. For example, the preset may already include the maximum nine effects blocks or may not have enough DSP horsepower to add another of the specific model type (see ["Dynamic DSP"](#)). In such a case, the PASTE BLOCK switch is grayed-out.

Clearing an Effect

1. Touch the stomp switch you want to clear and press ACTION.
Action switches appear around the selected effect.
2. Press the CLEAR BLOCK switch.

Adding a Block to Favorites

Once you've dialed in an effects block exactly the way you like it, you can save its settings as a "Favorite." This allows you to add it very quickly to other tones, complete with your stored parameter settings and customized bypass assignment (excluding any existing footswitch, controller, and snapshot assignments).

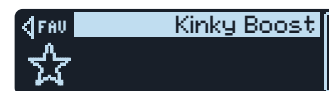
Favorites appear in the model menu's "Favorites" category and can be renamed, reordered, and cleared. (Favorites does not appear within the device's Category menu until you add at least one block as a favorite.) You can add up to 128 Favorites.

NOTE: Favorites are auto-assigned to a footswitch or EXP pedal when added to your preset, just as with non-Favorite model types.
Split, and Merge Mixer block types cannot be added to favorites.

1. While in Stomp or Signal Flow mode, choose one of your preferred effects and tweak its settings exactly how you like them, and with or without a custom footswitch assignment.
2. Press ACTION, press $\triangleright$, and then press the Add to Favorites footswitch.
3. To replace an existing favorite, select the favorite you want to replace, and press the Replace Favorite footswitch.*

To add a new favorite, select the location where you want to add it, and press the Add New Favorite footswitch.

The model is added to the Favorites Category list.



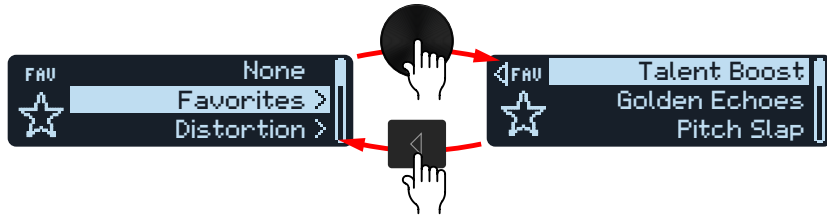
***NOTE:** If you replace and overwrite an existing favorite that is already in use within your presets, the presets remain unchanged. However, if you want to "update" any preset to use the newly-edited favorite, select the existing favorite block with the preset and turn the Big Knob one click away and then back to effectively "reload" the updated favorite.

TIP: Using the HX Edit app, you can export your favorites to your computer as .fav files, allowing you to build a virtually limitless collection that can be imported back into your device's Favorites library at any time—as well as shared between devices, or with other Helix/HX device and Helix Native owners! HX Edit also offers several features for easily managing your Favorites library.

4. Put your Favorites to use...

Within any preset, select an empty block location and turn the Big Knob clockwise. All your perfectly-tweaked favorite effects instantly appear as the first options, without having to open the model list at all! The switch's LED color reflects the selected favorite's model category.

Or, select an empty block location, press the Big Knob, and choose Favorites to see your list.



5. OPTIONAL: You can further customize your favorites list by opening the Model list again, choose the Favorites category, select any favorite in the list, and press ACTION.

- To reorder the selected favorite up or down the list, press Reorder Up or Reorder Down
- To rename the selected favorite, press Rename Favorite
- To clear the selected favorite, press Clear Favorite
- To clear all favorites, press Clear All Favorites. A dialog appears. Press OK to confirm
- Press to exit the Action options

Saving a Model's User Default Settings

If you find yourself continually re-tweaking your commonly-used models the same way, you can save a model's settings as its new defaults (excluding Split or Merge Mixer blocks' settings), so the model shows up that way every time.

1. Choose any model and tweak it exactly how you like it.

Configure any of its parameters and bypass state. (Existing MIDI, bypass, snapshot, and controller assignments are *not* saved with the User Model Defaults.)

2. Press ACTION, press and then press Model Default.

All your current parameter settings are saved as the defaults for future uses of the model. (Existing presets that include the model are not affected.)

NOTE: To return a model to factory defaults, press ACTION, press and then press Factory Default. No presets' model settings are changed—you'll see the factory defaults restored the next time you insert the model.

Customizing a Footswitch

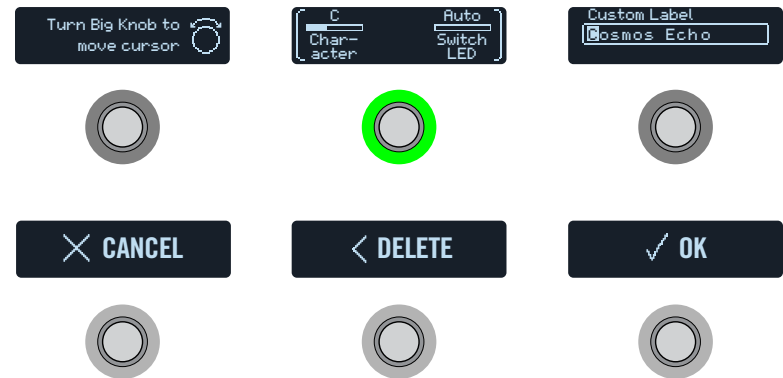
If you want to customize the text label displayed on a scribble strip, or change the footswitch's LED color, follow these steps.

1. Touch the stomp switch you want to customize and press ACTION.

Action switches appear around the selected effect.

2. Press the CUSTOMIZE switch.

The customize screens appear:



Turn the Big Knob to move the cursor.

Turn Knob 1 (Character) to change the selected character.

SHORTCUT: Press Knob 1 (Character) to cycle through upper case, lower case, 0, and [SPACE].

Press the DELETE switch to delete the selected character and shift all following characters to the left.

Delete all characters to remove the custom label, after which the footswitch label displays its normal assignment.

3. Turn Knob 3 (Switch LED) to select the desired color for the footswitch's LED ring (or turn it off).

Normally, you should leave this set to "Auto."

4. When finished, press the OK switch.

Preset Mode

Adding and editing effects is all well and good, but at that point, why not just get a bunch of separate pedals? One of the main advantages of boxes like the HX Effects unit is the ability to store many different “pedalboards”—each with potentially different effects, configurations, and settings—and recall them all with a single footswitch press. These pedalboards are saved as *presets*.

The HX Effects device stores up to 32 banks of four presets (A, B, C, and D) for a total of 128 presets.

If not already there, press MODE to select Preset mode.

Footswitches 1-6 are used for navigating banks and presets:



1. Press BANK ^ or BANK v to choose the desired bank.

The banks' presets flash, indicating they're ready to load.

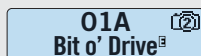
2. Press one of the four preset switches to load the preset.

NOTE: The HX Effects device can be set to return to Stomp mode after pressing a preset switch automatically. See [“Global Settings > Switches”](#)

NOTE: The current preset switch displays a small camera icon in the upper right corner. The number indicates the current snapshot. Snapshots offer a whole additional level of capabilities within each preset! See [“Snapshots”](#)

Alternatively, when in Preset mode, turn the Big Knob to select a preset.

NOTE: When the selected preset is in an edited state, a small “E” is displayed:

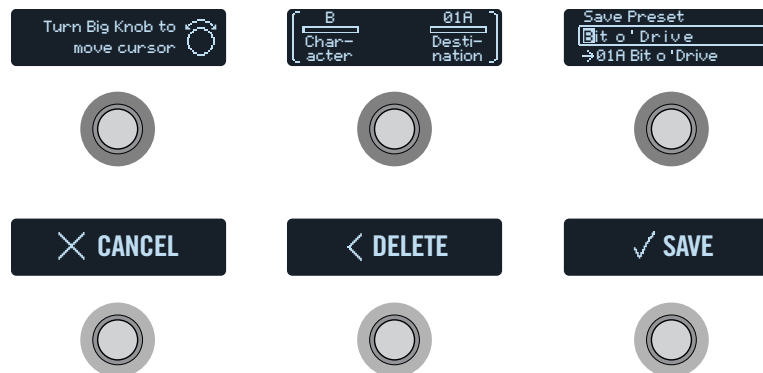


Saving/Naming a Preset



SHORTCUT: To quickly save any changes to the current preset, just press SAVE twice.

1. Press SAVE to open the Save Preset screens:



Turn the Big Knob to move the cursor.

Turn Knob 1 (Character) to change the selected character.



SHORTCUT: Press Knob 1 (Character) to cycle through A, a, 0, and [SPACE].

Press the DELETE switch to delete the selected character and shift all following characters to the left.

2. Turn Knob 3 (Destination) to choose the preset location you wish to overwrite.

You can overwrite any of the 128 presets.

3. When finished, press the SAVE switch.



TIP: You can use the HX Edit application to Export any presets from your HX Effects hardware - this is a great way to save unlimited backup copies of all your custom presets to your computer! Please see the current HX Edit Pilot's Guide for details, available from <https://line6.com/support/manuals/>



NOTE: HX Effects-exported preset files cannot be imported and used with Helix family devices (Helix Floor, Helix Rack and Helix LT) or with the Helix Native plugin software. Likewise, preset and setlist files exported from Helix family devices or Helix Native plugin are not able to be imported and used on HX Effects devices.

Creating/Restoring Full Device Backups

Creating full backups of your device's presets, IRs, favorites, custom user model defaults, and global settings is highly recommended to make sure your precious custom tones and configurations are never lost! Fortunately, the free Line 6 **HX Edit** app makes it easy to create full backups and restore them anytime (available from line6.com/software). Please refer to the [HX Edit Pilot's Guide](#) for steps on how to use its built-in Backup & Restore features.

TAP Tempo

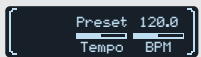
Press TAP/TUNER repeatedly to set the tempo in BPM (Beats Per Minute).

Certain Delay, Reverb, and Modulation blocks' parameters, such as Time, Rate, and Speed, can be represented with note values (1/4-note, dotted 1/8-note, etc.) or fixed numeric values (ms or Hz). When set to note values, the parameter will follow Tap Tempo. Press the parameter knob to toggle between note or ms (or Hz) values.

NOTE: The HX Effects device does not respond to pressing TAP at tempi lower than 40.0 BPM to ensure that multiple fast tap entries are recognized properly. You may still manually select a tempo as low as 20.0 BPM using the Global Settings or Knob 3 BPM parameter options described below.

Also, see the Tempo options available in "[Global Settings > MIDI/Tempo](#)" where you can fine-tune your Tap Tempo value or configure your HX Effects device to sync tempo with external MIDI hardware and software via MIDI Clock.

 **SHORTCUT:** Touch TAP/TUNER to briefly display Tempo parameters above Footswitch 6.



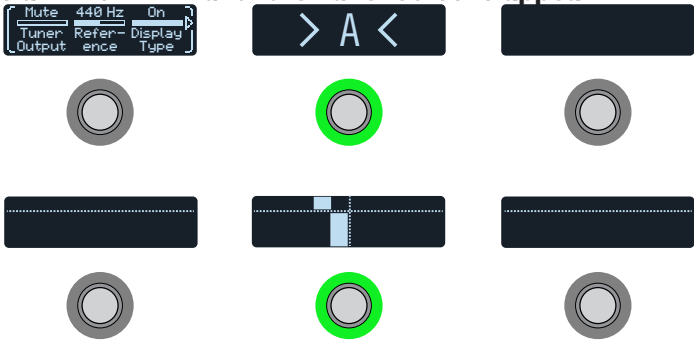


Knob	Parameter	Description
2	Tempo	Choose whether the tempo is stored and recalled with each snapshot, recalled with each preset, or is applied globally across all presets and snapshots.
3	BPM	Depending on the Knob 2 (Tempo) setting, this value is saved per snapshot, per preset, or globally.

Tuner

HX Stomp offers three Tuner types, Fine (shown below), Coarse, and Strobe, selectable from Knob 4 of the Tuner Settings—see the settings table below. To use the default Fine tuner type...

1. Press and Hold TAP until the Tuner screens appear.



2. Pluck an individual string on your guitar.

Your note is detected and displayed in the top center scribble strip, with a left arrow appearing when flat, a right arrow when sharp, and both arrows when in tune.

On the bottom row of scribble strips, when the moving indicator is left of center, your string is flat. When the indicator is right of center your string is sharp. When the indicator is displayed within the bottom left or right strip, you'll also see the footswitch LEDs light red to indicate you are flat or sharp, respectively.

When the indicator is displayed within the bottom center box, a small indicator appears on the top row for greater precision. You'll also see the center footswitch LEDs light yellow when close and green when in tune, as shown in the image above.

3. To exit the Tuner, step on any footswitch.

All tuner settings are global.

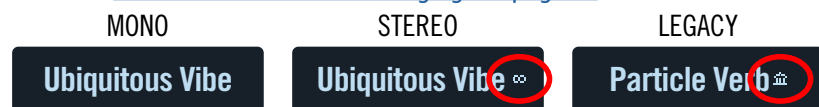
TIP: If you have a favorite pedal tuner and wish to use it instead of the tuner within your HX Effects device, connect Send 1 or 2 to your tuner's input and turn Knob 1 (Tuner Out) to select "Send12." This way, every time you hold the TAP footswitch, the HX Effects device will automatically route the signal to your favorite tuner.

Tuner Settings

Page	Knob	Parameter	Description
1	1	Tuner Output	Determines the active output while the Tuner screen is active. If you prefer to hear nothing while tuning, choose "Mute." Optionally, choose "L/R" to hear your unprocessed signal from the Main Left/Right outputs, or "Send12" to hear it from the Send 1/2 outputs.
	2	Reference	If you'd like to tune to a reference other than standard 440 Hz, select from 425 to 455 Hz.
	3	Display Type	Choose between three Tuner types: Fine (needle tuner with fine-tuning bar, the default type), Coarse (needle tuner with no fine-tuning bar), or Strobe. The Tuner accurately tunes notes down to 21.83 Hz (low F) for 5-string basses.
2	1	Offsets	When set to "On," enables the Tuner offsets displayed on Pages 2~4
	2	String6 Offset	Some guitarists feel that tuning certain strings slightly sharp or flat in relation to concert tuning can improve intonation. String offsets calibrate the tuner so that these slightly out-of-tune pitches appear as in tune. String 6 is low E, and String 1 is high E. Tuning offsets won't be applied unless Knob 1 (Offsets) on Page 2 is turned on.
	3	String5 Offset	
3	1	String4 Offset	
	2	String3 Offset	
	3	String2 Offset	
4	1	String1 Offset	

Effects

The HX Effects device includes all the effects of the Line 6 award-winning Helix guitar processors. These are found in the Mono and Stereo subcategories in the model list. In addition, the HX Effects unit includes selected effects from the classic DL4, DM4, MM4, and FM4 stompboxes as well as M13, M9, and M5 processors; these appear in the Legacy subcategory. Some Legacy models process the signal in mono, and others in stereo. See [“Block Order and Stereo Imaging” on page 40](#) for more info.



From Stomp mode, touch an effect's footswitch and turn the Big Knob to change its model, or press the Big Knob to enter the Model menu.



TIP: Once you've added one or more models as “Favorites,” you'll see Favorites appear as the first category within the Model menu—see [page 14](#).

Distortion Models

Model	Subcategories	Based On*
Kinky Boost	Mono, Stereo	Xotic® EP Booster
Deranged Master	Mono, Stereo	Dallas Rangemaster Treble Booster
Minotaur	Mono, Stereo	Klon® Centaur
Teemah!	Mono, Stereo	Paul Cochrane Timmy® Overdrive
Heir Apparent	Mono, Stereo	Analogman Prince of Tone
Tone Sovereign	Mono, Stereo	Analogman King of Tone
Alpaca Rogue	Mono, Stereo	Way Huge® Red Llama (modded)
Compulsive Drive	Mono, Stereo	Fulltone® OCD
Dhyana Drive	Mono, Stereo	Hermida Zendrive
Horizon Drive	Mono, Stereo	Horizon Precision Drive
Valve Driver	Mono, Stereo	Chandler Tube Driver
Top Secret OD	Mono, Stereo	DOD® OD-250
Prize Drive	Mono, Stereo	Nobels® ODR-1(bc)
Scream 808	Mono, Stereo	Ibanez® TS808 Tube Screamer®

Model	Subcategories	Based On*
Pillars	Mono, Stereo	Earthquaker Devices® Plumes
Hedgehog D9	Mono, Stereo	MAXON® SD9 Sonic Distortion
Stupor OD	Mono, Stereo	BOSS® SD-1 Overdrive
Deez One Vintage	Mono, Stereo	BOSS DS-1 Distortion (Made-in-Japan)
Deez One Mod	Mono, Stereo	BOSS DS-1 Distortion (Keeley modded)
Ratatoullie Dist	Mono, Stereo	Pro Co RAT (with LM308 opamp)
Vermin Dist	Mono, Stereo	Pro Co RAT
Vital Dist	Mono, Stereo	Earthquaker Devices Life - Octave/Distortion circuit
Vital Boost	Mono, Stereo	Earthquaker Devices Life - Boost circuit
KWB	Mono, Stereo	Benadrian Kowloon Walled Bunny Distortion
Legendary Drive	Mono, Stereo	Carvin® VLD1 Legacy Drive (hi gain channel)
Swedish Chainsaw	Mono, Stereo	BOSS HM-2 Heavy Metal Distortion (MIJ)
Arbitrator Fuzz	Mono, Stereo	Arbiter® Fuzz Face®
Pocket Fuzz	Mono, Stereo	Jordan Boss Tone Fuzz
Bighorn Fuzz	Mono, Stereo	'73 Electro-Harmonix® Ram's Head Big Muff Pi
Triangle Fuzz	Mono, Stereo	Electro-Harmonix Big Muff Pi
Dark Dove Fuzz	Mono, Stereo	Electro-Harmonix Russian Big Muff Pi
Ballistic Fuzz	Mono, Stereo	Euthymia ICBM Fuzz
Industrial Fuzz	Mono, Stereo	Z.Vex Fuzz Factory
Tycoctavia Fuzz	Mono, Stereo	Tycobrahe® Octavia
Wringer Fuzz	Mono, Stereo	Garbage's modded BOSS FZ-2
Thrifter Fuzz	Mono, Stereo	Line 6 Original
Xenomorph Fuzz	Mono, Stereo	Subdecay Harmonic Antagonizer
Megaphone	Mono, Stereo	Megaphone
Bitcrusher	Mono, Stereo	Line 6 Original
Ampeg Scrambler	Mono, Stereo	Ampeg® Scrambler Bass Overdrive
ZeroAmp Bass DI	Mono, Stereo	Tech 21® SansAmp Bass Driver DI V1
Regal Bass DI	Mono, Stereo	Noble Preamp Bass DI
Obsidian 7000	Mono, Stereo	Darkglass® Electronics Microtubes® B7K Ultra
Clawthorn Drive	Mono, Stereo	Wounded Paw Battering Ram

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Model	Subcategories	Based On*
Tube Drive	Legacy	Chandler Tube Driver
Screamer	Legacy	Ibanez Tube Screamer
Overdrive	Legacy	DOD Overdrive/Preamp 250
Classic Dist	Legacy	Pro Co RAT
Heavy Dist	Legacy	BOSS Metal Zone
Colordrive	Legacy	Colorsound® Overdriver
Buzz Saw	Legacy	Maestro® Fuzz Tone
Facial Fuzz	Legacy	Arbiter Fuzz Face
Jumbo Fuzz	Legacy	Vox® Tone Bender
Fuzz Pi	Legacy	Electro-Harmonix Big Muff Pi
Jet Fuzz	Legacy	Roland® Jet Phaser
L6 Drive	Legacy	Colorsound Overdriver (modded)
L6 Distortion	Legacy	Line 6 Original
Sub Oct Fuzz	Legacy	PAiA Roctave Divider
Octave Fuzz	Legacy	Tycobrahe Octavia
Bronze Master	Legacy	Maestro Bass Brassmaster
Killer Z	Legacy	BOSS Metal Zone MT-2

Dynamics Models

Model	Subcategories	Based On*
Deluxe Comp	Mono, Stereo	Line 6 Original
Red Squeeze	Mono, Stereo	MXR Dyna Comp
Kinky Comp	Mono, Stereo	Xotic SP Compressor
Ampeg Opto Comp	Mono, Stereo	Ampeg Opto Comp Compressor
Rochester Comp	Mono, Stereo	Ashly® CLX-52 (in conjunction w/ B. Sheehan)
LA Studio Comp	Mono, Stereo	Teletronix® LA-2A®
3-Band Comp	Mono, Stereo	Line 6 Original
Noise Gate	Mono, Stereo	Line 6 Original
Hard Gate	Mono, Stereo	Line 6 Original
Horizon Gate	Mono, Stereo	Horizon Precision Drive - Gate Circuit

Model	Subcategories	Based On*
Autoswell	Mono, Stereo	Line 6 Original
Feedbacker †	Mono	Line 6 Original, Feedback Generator
Tube Comp	Legacy	Teletronix LA-2A
Red Comp	Legacy	MXR Dyna Comp
Blue Comp	Legacy	BOSS CS-1
Blue Comp Treb	Legacy	BOSS CS-1 (Treble switch on)
Vetta Comp	Legacy	Line 6 Original
Vetta Juice	Legacy	Line 6 Original
Boost Comp	Legacy	MXR Micro Amp

EQ Models

Model	Subcategories	Based On*
Simple EQ	Mono, Stereo	Line 6 Original
Low and High Cut	Mono, Stereo	Line 6 Original
Low/High Shelf	Mono, Stereo	Line 6 Original
Parametric	Mono, Stereo	Line 6 Original
Tilt	Mono, Stereo	Line 6 Original
10 Band Graphic	Mono, Stereo	MXR 10-Band Graphic EQ
Cali Q Graphic	Mono, Stereo	MESA/Boogie Mark IV Graphic EQ
Acoustic Sim	Mono, Stereo	Line 6 Original

Modulation Models

Model	Subcategories	Based On*
Optical Trem	Mono, Stereo	Fender® optical tremolo circuit
60s Bias Trem	Mono, Stereo	Vox AC-15 Tremolo
Tremolo/Autopan	Mono, Stereo	BOSS PN-2
Harmonic Tremolo	Mono, Stereo	Line 6 Original
Bleat Chop Trem	Mono, Stereo	Lightfoot Labs Goatkeeper
Script Mod Phase	Mono, Stereo	MXR Phase 90
Pebble Phaser	Mono, Stereo	Electro-Harmonix Small Stone
Ubiquitous Vibe	Mono, Stereo	Shin-ei Uni-Vibe®

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†The Feedbacker model utilizes polyphonic pitch shifting and, therefore, is extremely DSP-intensive—possibly utilizing up to half of all DSP available for Path 1 or Path 2!

Model	Subcategories	Based On*
FlexoVibe	Mono, Stereo	Line 6 Original
Deluxe Phaser	Mono, Stereo	Line 6 Original
Gray Flanger	Mono, Stereo	MXR 117 Flanger
Harmonic Flanger	Mono, Stereo	A/DA Flanger
Courtesan Flange	Mono, Stereo	Electro-Harmonix Deluxe EM
Dynamix Flanger	Mono, Stereo	Line 6 Original
Chorus	Mono, Stereo	Line 6 Original
70s Chorus	Mono, Stereo	BOSS CE-1
PlastiChorus	Mono, Stereo	Modded Arion SCH-Z chorus
Ampeg Liquifier Chorus	Mono, Stereo	Ampeg Liquifier Chorus
Trinity Chorus	Stereo	Dytronics® Tri-Stereo Chorus
4-Voice Chorus	Mono, Stereo	Line 6 Original
Bubble Vibrato	Mono, Stereo	BOSS VB-2 Vibrato
Vibe Rotary	Stereo	Fender Vibratone
122 Rotary	Stereo	Leslie® 122
145 Rotary	Stereo	Leslie 145
Triple Rotary	Mono, Stereo	Yamaha® RA-200
Retro Reel	Mono, Stereo	Line 6 Original
Double Take	Mono, Stereo	Line 6 Original
Poly Detune †	Mono	Line 6 Original
AM Ring Mod	Mono, Stereo	Line 6 Original
Pitch Ring Mod	Stereo	Line 6 Original
Pattern Tremolo	Legacy	Line 6 Original
Panner	Legacy	Line 6 Original
Bias Tremolo	Legacy	1960 Vox AC-15 Tremolo
Opto Tremolo	Legacy	1964 Fender Deluxe Reverb®
Script Phase	Legacy	MXR Phase 90 (script logo version)
Panned Phaser	Legacy	Ibanez Flying Pan
Barberpole	Legacy	Line 6 Original
Dual Phaser	Legacy	Mu-Tron® Bi-Phase
U-Vibe	Legacy	Shin-ei Uni-Vibe

Model	Subcategories	Based On*
Phaser	Legacy	MXR Phase 90
Pitch Vibrato	Legacy	BOSS VB-2
Dimension	Legacy	Roland Dimension D
Analog Chorus	Legacy	BOSS CE-1
Tri Chorus	Legacy	Dytronics Tri-Stereo Chorus
Analog Flanger	Legacy	MXR Flanger
Jet Flanger	Legacy	A/DA Flanger
AC Flanger	Legacy	MXR Flanger
80A Flanger	Legacy	A/DA Flanger
Frequency Shift	Legacy	Line 6 Original
Ring Modulator	Legacy	Line 6 Original
Rotary Drum	Legacy	Fender Vibratone
Rotary Drum/Horn	Legacy	Leslie 145
Tape Eater	Legacy	Line 6 Original
Warble-Matic	Legacy	Line 6 Original
Random S&H	Legacy	Line 6 Original
Sweeper	Legacy	Line 6 Original

Delay Models

Model	Subcategories	Based On*
Simple Delay	Mono, Stereo	Line 6 Original
Mod/Chorus Echo	Mono, Stereo	Line 6 Original
Dual Delay	Stereo	Line 6 Original
Multitap 4	Stereo	Line 6 Original
Multitap 6	Stereo	Line 6 Original
Ping Pong	Stereo	Line 6 Original
Sweep Echo	Mono, Stereo	Line 6 Original
Ducked Delay	Mono, Stereo	TC Electronic® 2290
Reverse Delay	Mono, Stereo	Line 6 Original
Vintage Digital	Mono, Stereo	Line 6 Original
Vintage Swell	Mono, Stereo	Line 6 Original

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† The Poly Detune model utilizes polyphonic pitch shifting and, therefore, is extremely DSP-intensive.

Model	Subcategories	Based On*
Pitch Echo	Mono, Stereo	Line 6 Original
Transistor Tape	Mono, Stereo	Maestro Echoplex® EP-3
Cosmos Echo	Mono, Stereo	Roland RE-201 Space Echo®
Harmony Delay	Stereo	Line 6 Original
Bucket Brigade	Mono, Stereo	BOSS DM-2
Adriatic Delay	Mono, Stereo	BOSS DM-2 w/ Adrian Mod
Adriatic Swell	Mono, Stereo	Line 6 Original
Elephant Man	Mono, Stereo	Electro-Harmonix Deluxe Memory Man
Multi Pass	Mono, Stereo	Line 6 Original
Heliosphere	Mono, Stereo	Line 6 Original
Poly Sustain †	Mono	Line 6 Original
Glitch Delay	Mono, Stereo	Line 6 Original
Euclidean Delay	Mono, Stereo	Line 6 Original
ADT	Mono, Stereo	Line 6 Original
Crisscross	Mono, Stereo	Line 6 Original
Tesselator	Mono, Stereo	Line 6 Original
Ratchet	Mono, Stereo	Line 6 Original
Ping Pong	Legacy	TC Electronic 2290
Dynamic	Legacy	TC Electronic 2290
Stereo	Legacy	Line 6 Original
Digital	Legacy	Line 6 Original
Dig w/Mod	Legacy	Line 6 Original
Reverse	Legacy	Line 6 Original
Lo Res	Legacy	Line 6 Original
Tube Echo	Legacy	Maestro Echoplex EP-1
Tape Echo	Legacy	Maestro Echoplex EP-3
Sweep Echo	Legacy	Line 6 Original
Echo Platter	Legacy	Binson® EchoRec®
Analog Echo	Legacy	BOSS DM-2
Analog w/Mod	Legacy	Electro-Harmonix Deluxe Memory Man
Auto-Volume Echo	Legacy	Line 6 Original

Model	Subcategories	Based On*
Multi-Head	Legacy	Roland RE-101 Space Echo
Bubble Echo	Legacy	Line 6 Original
Phaze Eko	Legacy	Line 6 Original

Reverb Models

Model	Subcategory	Based On*
Dynamic Hall	Mono, Stereo	Line 6 Original
Dynamic Plate	Mono, Stereo	Line 6 Original
Dynamic Room	Mono, Stereo	Line 6 Original
Dynamic Ambience	Mono, Stereo	Line 6 Original
Dynamic Bloom	Mono, Stereo	Line 6 Original
Shimmer	Mono, Stereo	Line 6 Original
Hot Springs	Mono, Stereo	Line 6 Original
Nonlinear	Mono, Stereo	Line 6 Original
Glitz	Mono, Stereo	Line 6 Original
Ganymede	Mono, Stereo	Line 6 Original
Searchlights	Mono, Stereo	Line 6 Original
Plateaux	Mono, Stereo	Line 6 Original
Double Tank	Mono, Stereo	Line 6 Original
Plate	Legacy	Line 6 Original
Room	Legacy	Line 6 Original
Chamber	Legacy	Line 6 Original
Hall	Legacy	Line 6 Original
Echo	Legacy	Line 6 Original
Tile	Legacy	Line 6 Original
Cave	Legacy	Line 6 Original
Ducking	Legacy	Line 6 Original
Octo	Legacy	Line 6 Original
'63 Spring	Legacy	Line 6 Original
Spring	Legacy	Line 6 Original
Particle Verb	Legacy	Line 6 Original

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† The Poly Sustain model utilizes polyphonic pitch shifting and, therefore, is extremely DSP-intensive.

Pitch/Synth Models

Model	Subcategories	Based On*
Pitch Wham	Mono, Stereo	Digitech Whammy®
Twin Harmony	Mono, Stereo	Eventide® H3000
Simple Pitch	Mono, Stereo	Line 6 Original
Dual Pitch	Mono, Stereo	Line 6 Original
Boctaver	Mono/Stereo	Boss OC-2 Octaver
3 OSC Synth	Stereo	Line 6 Original
Poly Pitch †	Mono	Line 6 Original
Poly Wham †	Mono	Line 6 Original
Poly Capo †	Mono	Line 6 Original
12 String †	Mono	Line 6 Original
3 Note Generator ‡	Mono, Stereo	Line 6 Original
4 OSC Generator ‡	Mono, Stereo	Line 6 Original
Bass Octaver	Legacy	EBS® OctaBass
Smart Harmony	Legacy	Eventide H3000
Octi Synth	Legacy	Line 6 Original
Synth O Matic	Legacy	Line 6 Original
Attack Synth	Legacy	Korg® X911 Guitar Synth
Synth String	Legacy	Roland GR700 Guitar Synth
Growler	Legacy	Line 6 Original
Buzz Wave	Legacy	Line 6 Original
Rez Synth	Legacy	Line 6 Original
Seismik Synth	Legacy	Line 6 Original
Analog Synth	Legacy	Line 6 Original
Synth Lead	Legacy	Line 6 Original
String Theory	Legacy	Line 6 Original
Synth FX	Legacy	Line 6 Original
Saturn 5 Ring Mod	Legacy	Line 6 Original
Synth Harmony	Legacy	Line 6 Original
Double Bass	Legacy	Line 6 Original

Filter Models

Model	Subcategories	Based On*
Mutant Filter	Mono, Stereo	Musitronics Mu-Tron III
Mystery Filter	Mono, Stereo	Korg A3
Autofilter	Mono, Stereo	Line 6 Original
Asheville Pattrn	Mono, Stereo	Moog® Moogerfooger® MF-105M MuRF Filter
Voice Box	Legacy	Line 6 Original
V Tron	Legacy	Musitronics® Mu-Tron III
Q Filter	Legacy	Line 6 Original
Seeker	Legacy	Z Vex Seek Wah
Obi Wah	Legacy	Oberheim® voltage-controlled S&H filter
Tron Up	Legacy	Musitronics Mu-Tron III (up position)
Tron Down	Legacy	Musitronics Mu-Tron III (down position)
Throbber	Legacy	Electrix® Filter Factory
Slow Filter	Legacy	Line 6 Original
Spin Cycle	Legacy	Craig Anderton's Wah/Anti-Wah
Comet Trails	Legacy	Line 6 Original

Wah Models

Model	Subcategories	Based On*
UK Wah 846	Mono, Stereo	Vox V846
Teardrop 310	Mono, Stereo	Dunlop® Cry Baby® Fasel model 310
Fassel	Mono, Stereo	Dunlop Cry Baby Super
Weeper	Mono, Stereo	Arbiter Cry Baby
Chrome	Mono, Stereo	Vox V847
Chrome Custom	Mono, Stereo	Modded Vox V847
Throaty	Mono, Stereo	RMC® Real McCoy 1
Vetta Wah	Mono, Stereo	Line 6 Original
Colorful	Mono, Stereo	Colorsound Wah-fuzz
Conductor	Mono, Stereo	Maestro Boomerang
Teardrop Bass Q	Mono, Stereo	Modified Dunlop 105Q

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† These indicated Pitch/Synth models utilize polyphonic pitch shifting and, therefore, are extremely DSP-intensive. Each one might take up to half of a preset's available DSP!

‡ The 3 Note Generator and 4 OSC Generator models make sound without any instrument connected, therefore, their blocks are turned off by default. Be sure to first set your Helix volume to low, and press BYPASS to turn them on.

Volume/Pan Models

Model	Subcategories	Based On*
Volume Pedal	Mono, Stereo	Line 6 Original
Gain	Mono, Stereo	Line 6 Original
Pan	Stereo	Line 6 Original
Stereo Width	Stereo	Line 6 Original
Stereo Imager	Stereo	Line 6 Original

Common FX Settings

Parameter	Description
Drive	Adjusts the amount of overdrive, distortion, or fuzz.
Bass	Adjusts the bass level.
Mid	Adjusts the midrange level.
Treble	Adjusts the treble level.
Speed	Adjusts the speed of the effect, with higher settings providing faster rates. Press the knob to toggle between Hz and note values. Choosing a Hz value provides a specific modulation speed in cycles per second; choosing a note value provides a time based on the current tempo. Not all Speed parameters can be synced to note values, as they may be non-linear and highly interactive. (Note that stepping on TAP once resets any LFO-based effects, such as tremolos and rotary speakers.)
Rate	Adjusts the rate of the effect, with higher settings providing faster rates. Press the knob to toggle between numeric and note values. Not all Rate parameters can be synced to note values, as they may be non-linear and highly interactive. (Note that stepping on TAP once resets any LFO-based effects, such as tremolos and rotary speakers.)
Time	Adjusts the delay/repeat time, with higher settings providing longer delays. Press the knob to toggle between ms and note values. Choosing a ms value provides a specific time in milliseconds; choosing a Note Division value provides a time based on the current tempo. With a note division value, this parameter's value is retained when changing models.
Depth	Adjusts the intensity of the modulation. Higher settings result in more extreme pitch bending, wobble, or throb, depending on the effect.
Feedbk	Adjusts the amount of delayed signal fed back into the effect. Higher settings can provide more dramatic textures.
Decay	Sets the length of time the reverb effect sustains.
Predly	Determines predelay—or the time before the reverb effect is heard.

Parameter	Description
Scale	On stereo delays, the Scale offers control over the left & right channel repeats proportionately. The left channel repeats following the Time value, and the right channel will repeat at a time that is the percentage of the left time. For example, if a delay's Time is set for 1 second and the Scale set to 75%, the left channel will repeat at 1 second and the right at 750 milliseconds (ms).
Spread	Spread differs slightly among stereo delay effects. For most delay, it adjusts how widely the repeats bounce left and right. For example, with the Ping Pong Delay, 0 is in the middle (mono), and 10 is full left to right panning on the repeats. For modulated stereo delays, Spread affects the LFOs' (low frequency oscillators) stereo modulation behavior. At 0, the LFOs are in sync. At 10, the two LFOs are 180 degrees out of sync, so that when one side is modulating up, the other side is modulating down.
Headrm	Some mod and delay pedals' internal signal paths exhibit a bit of grit, especially when placed after a high-gain distortion block. Negative values increase the perceived amount of grit; positive values clean things up a bit. At 0dB, the model behaves like the original pedal.
Low Cut	Filters a portion of the block's bass and/or treble frequencies, which can help remove rumble and/or high-end harshness.
Hi Cut	
Mix	Blends the effected "wet" signal vs. the "dry" signal passed through the block. When set to 0%, the path bypasses the effect completely. When set to 100%, the entire path is fed through the effect, and no dry thru signal is heard.
Level	Adjusts the overall output level of the effects block. Be careful not to boost this parameter too high on multiple blocks, as digital clipping could occur. You should typically leave this at 0.0dB for most blocks. Where the original pedal's level or volume knob behavior doesn't really apply to dB values, 0.0-10 may be used.
Trails	<i>Trails Off:</i> Any delay repeats or reverb decays are instantly muted when the block is bypassed. <i>Trails On:</i> Any delay repeats or reverb decays continue to decay naturally when the block is bypassed or a different snapshot is selected.

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Impulse Response (IR)

Impulse Responses are mathematical functions representing the sonic measurements of specific audio systems (for the HX Effects device, acoustic guitar body resonance or speaker cabinet and microphone combinations). The HX Effects device can load and store up to 128 custom or third-party IRs at a time.



Single and Dual IR Blocks

From the IR menu, you can select either a mono **Single IR** block type or a stereo **Dual IR** block type.

- A Single IR block allows you to choose either the 1024-Sample or 2048-Sample higher resolution IR type. Optionally, you can add up to two Single 1024 IR blocks per preset (such as one each within the A & B parallel paths).*
- A Dual IR block provides the ability to load two of your IRs into one block, which can be independently set with different parameter settings, delayed, and panned as desired. You can add one Dual IR block max. per preset.*

IR Block Parameters

Individual IR Parameters - The following are available for an IR block. You'll see two sets of these parameters in a Dual IR block—one set for each of the block's "A" and "B" IRs.

Parameter	Description
IR Select	Loads the IR index 1-128 from the IR Library—see the following section.
Low Cut	Filters a portion of the IR block's bass/treble frequencies, which can help remove rumble and/or high-end harshness.
High Cut	
Mix	Adjusts the balance of dry and wet (IR processed) signals. 0% = completely dry, and 100% = completely wet.
Level	Adjusts the overall output level of the IR.
Dual IR Block Parameters - The following are additionally available for any Dual IR block.	
Pan	Pan the output of each IR independently up to 100% Left or 100% Right.
Polarity	Sets the polarity (Normal or Inverted) for either IR. If you encounter audible phasing, try setting one IR's Polarity to Inverted.
Dual IR Block "Both" Parameters - The following are applied to both IRs in a Dual IR block.	
Delay	Delays either the "A" or "B" IR up to 50 ms. You can use this to correct minor phasing issues between the blocks' IRs (as an alternative to using Polarity), or to delay the output of one IR to simulate a "double tracking" effect—most effective when the two IRs are also panned opposite.
Mix	Adjusts the balance of dry and wet (IR processed) signals for the entire Dual IR block. 0% = completely dry, and 100% = completely wet.

IR Formats

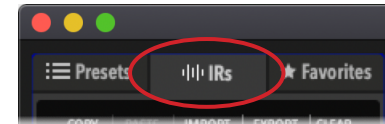
For all Helix/HX family devices, as well as Helix Native plugin, the following IR file types are supported.

- **Helix Impulse Response (.hir)** - This is the proprietary Line 6 IR file format. All IRs purchased through the Line 6 Marketplace are of this file type—see ["Marketplace" on page 5](#).
- **WAV (.wav)** - Most IR producers and vendors typically offer IRs within the .wav format. It is possible to import a mono or stereo .wav file IR, regardless of its bit depth, length, or sample rate frequency. Upon import, IR .wav files are automatically converted to 48kHz sample rate, 32-bit, mono, and 2,048 sample length for use by HX Effects. You can use the HX Edit app's Preferences to set the preferred behavior for importing Stereo WAV IRs (import the left or right channel data of the file, or a mix of both).

Loading IRs into Helix

Loading Impulse Response files into HX Effects requires connecting to the HX Edit application in your Mac or Windows computer. The HX Edit application is available as a free download from line6.com/software.

1. **Connect HX Effects to your computer via USB and open the HX Edit application.**
2. **Click the IRs tab.**



3. **Drag one or more IR files from the desktop or any Finder window directly into the HX Edit application's IRs list (or use the HX Edit IR - Import command).**

The HX Edit app updates the HX Effects hardware's IR list automatically. HX Effects can load and store up to 128 IRs at a time.

Loading an IR in an IR Block

Once you've imported IR files into HX Effects (see previous section), you can then put them to use by adding one or more IR blocks into your tone.

1. **Turn the Big Knob to select an empty block location.**
2. **Push the Big Knob to open the model menu, turn the Big Knob to select the "IR" category. You can choose either a Single IR, for mono operation, or a Dual IR.**

NOTE: When choosing a Single IR, you may optionally choose the higher fidelity 2048-sample IR (there is a max. limit of one 2048-sample IR block per Path, or two 1024-sample IR blocks per path).

NOTE: You may optionally choose the higher fidelity 2048-sample IR (there is a max. limit of one 2048-sample IR block or two 1024-sample IR blocks per preset).

3. Turn Knob 1 (IR Select) to choose the IR index (1~128) to utilize the preferred IR file within your IR library.

Use the other knobs on this page and the next to fine-tune the IR settings to taste—see the following Settings table.

TIP: Optionally, you can create a Snapshot assignment (see [page 32](#)) on the IR Select parameter, which then allows you to change between different IRs within the block per snapshot!

IR File Reference

Once an IR block is configured to utilize an IR index slot that includes an imported IR (.wav or .hir) file, and the preset then saved, the preset creates a “reference signature” to the imported IR from its file name. Likewise, if you save your IR block as a Favorite (see [page 14](#)), or set the IR block’s current settings as the User Model Defaults (see [page 15](#)), the IR block also creates a reference to the specific IR file within your IR library. Therefore, if you reorder IRs within the IR library list, your preset (or IR Favorite or User Model Default block) will still intelligently reference the originally associated IR file, even if it now resides in a different IR Library index location. This also makes it easy to share your presets with friends since they can then insert the same IR file in any slot within their HX Effects IR library, and the preset will find it. Please also refer to the [HX Edit Pilot's Guide](#) for additional info regarding the use of IRs.

Send/Return

Each of the Send and Return jacks can be used independently or used in combination as a mono or stereo FX loop.



FX loops let you dynamically insert your favorite external stompboxes (or rack effects) into any location in your preset.

NOTE: Each Send and Return pair can be set for Instrument level (for inserting stompboxes) or Line level operation. See [“Global Settings > Ins/Outs”](#)

Send Settings

Knob	Parameter	Description
1	Send	Adjusts the level sent to your external devices.
2	Dry Thr	Adjusts the signal's level through the Send block, independent of the Knob 1 (Send) level. Normally, this should be set to 0.0dB.

Return Settings

Knob	Parameter	Description
1	Return	Adjusts the level received at the Return jack.
2	Mix	Blends the Return signal vs. the dry signal passed through the Return block. When set to 0%, the path bypasses the Return completely. When set to 100%, the entire signal is fed from the Return, and no dry thru signal is heard.

FX Loop Settings

Page	Knob	Parameter	Description
	1	Send	Adjusts the level sent to your external device.
	2	Return	Adjusts the level received at the Return jack.
1	3	Mix	Blends the FX loop signal vs. the dry signal passed through the FX Loop block. When set to 0%, the path bypasses the FX loop completely. When set to 100%, the entire path is fed through the FX loop, and no dry thru signal is heard. (This means that if you don't currently have your external gear connected to the Send & Return jacks, you won't hear any output signal unless you bypass the FX Loop block!)
2	1	Trails	<i>Trails Off:</i> An external stompbox would be instantly muted when the FX Loop block is bypassed. <i>Trails On:</i> An external delay or reverb stompbox would continue to decay naturally when the FX Loop block is bypassed, or a different snapshot is selected.

Looper

HX Effects offers three different Looper types: the 6 Switch, 1 Switch, and Shuffling Looper. You can add one Looper block per preset maximum.



Looper Models (Mono & Stereo)

Model	Based On*	Max. Loop Length	
		Half Speed	Full Speed
6 Switch Looper (Mono)	Line 6 Original	120 seconds	60 seconds
1 Switch Looper (Mono)	Line 6 Original	120 seconds	60 seconds
Shuffling Looper (Mono)	Line 6 Original	n/a	60 seconds
6 Switch Looper (Stereo)	Line 6 Original	60 seconds	30 seconds
1 Switch Looper (Stereo)	Line 6 Original	60 seconds	30 seconds
Shuffling Looper (Stereo)	Line 6 Original	n/a	30 seconds

! IMPORTANT! You may change presets while looping, but loop playback will stop unless the preset you select includes the *same* type of looper block (same 6 Switch, 1 Switch, or Shuffle looper type, and same mono or stereo type).



TIP: The **Command Center - HX Looper** options allow you to customize your Stomp Mode footswitches further to easily trigger multiple looper functions for the 6 Switch or 1 Switch Loopers—see [“Command Center” on page 43](#).

You can also control the 6 Switch or 1 Switch Loopers via external MIDI—see [“MIDI” on page 52](#).

6 Switch and 1 Switch Looper Settings

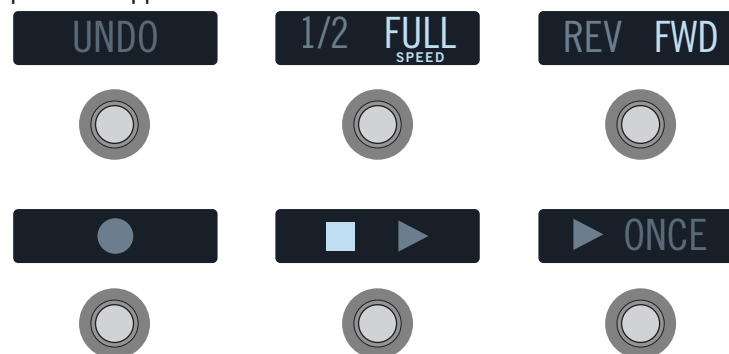
Page	Knob	Parameter	Description
	1	Playbk	Adjusts looper playback level. You may find it useful to turn this down a bit so your live guitar can be slightly louder.
1	2	Ovrdub	<i>Relatively</i> sets the level of your loop while overdubbing. For example, if your Overdub Level is set to 90%, each time your loop repeats, its volume will be reduced by 10%, sounding quieter and quieter with each overdub pass.
	3	Low Cut	Filters a portion of the loop's bass and/or treble frequencies, which can improve the mix with your live guitar.
2	1	High Cut	

Using the 6 Switch Looper

Once a 6 Switch Looper is added, pressing its Stomp mode footswitch activates its Looper Footswitch mode, where all its functions are conveniently laid out and labeled on the HX Effects footswitches.

1. From Stomp mode, press the footswitch labeled “6 Switch Looper.”

Looper mode appears:



Switch	Description
	Step on  to start recording a loop. Step on   to end the loop and immediately start playback. Step on  to overdub additional parts. Step on   again to stop playback.
UNDO	If you make a mistake on your last overdub, step on UNDO to erase it.
 ONCE	Step on  ONCE to play the recorded loop once through.
1/2 FULL SPEED	Recording at full speed and then switching to 1/2 speed will also drop the loop down one octave. Recording at 1/2 speed will double your looping memory, and switching to full speed will cause the loop to play at double speed (up an octave).
REV FWD	Step on REV/FWD to hear your loop backward.

 **NOTE:** If you press  while loop playback is stopped, this will always record a new loop, and any previous recording will be discarded.

 **IMPORTANT!** You may change presets while looping, but loop playback will stop unless the preset you select includes the same type of looper block (mono or stereo).

2. To return to the previous mode, press MODE (EDIT/EXIT).

Using the 1 Switch Looper

Unlike the 6 Switch Looper, the 1 Switch Looper does not utilize a special Looper Footswitch mode. As its name implies, all functions for the 1 Switch Looper are available from its assigned Stomp mode footswitch, described as follows.

1. Add a 1 Switch Looper block to your preset.

The Looper should be automatically assigned to one of your Stomp mode footswitches.

2. In Stomp footswitch mode, press the 1 Switch Looper switch.

The LED lights red, indicating the loop is recording.

3. Press the 1 Switch Looper switch again.

The LED lights green, indicating the loop is playing back.

4. Press the 1 Switch Looper switch again.

The LED lights amber, indicating the loop is in overdub mode. Subsequent presses of the switch toggle between the play and overdub modes.

5. While the Looper is in play or overdub mode, press and hold the switch for 1 second.

Your most recent recording is undone. Hold the switch again to redo the recording.

6. Quickly double-press the 1 Switch Looper switch.

Playback/recording stops, and the LED lights white, indicating a loop is in memory.

7. While Looper playback/recording is stopped, press and hold the switch for 1 second.

Your most recent recording is deleted, and the LED lights dim white.

Using the Shuffling Looper

Part looper, part sampler, part performance instrument—the Shuffling Looper chops up your recorded loop, randomizes the slices, and gives you control over reordering, octave shifting, reversing, repeating, and more.

1. Add a Shuffling Looper block to your preset.

The Looper should be automatically assigned to one of your Stomp mode footswitches.

2. Turn Knob 1 (Slices) to set the number of slices your loop will be chopped into.

3. Press the switch to begin recording.

The LED lights red, indicating the loop is recording.

4. When done recording your loop, press the switch.

The LED lights green, and the sliced loop sequence immediately plays.

5. During playback, adjust the following Shuffling Looper parameters (or assign them to controllers, like expression pedals):

- **Slices**—Determines the number of slices your loop will be chopped into
- **SeqLen**—Determines the number of steps in the sequence before it loops
- **Shuffle**—Determines the likelihood of slices shuffling/reordering
- **Pitch**—Determines the likelihood of your slices changing pitch, based on the new Interval 1 and Interval 2 settings
- **Reverse**—Determines the likelihood of slices playing backward
- **Repeat**—Determines the likelihood of slices repeating
- **Smooth**—Higher values apply smoothing between slices and can give a synth-pad type quality. Lower values maintain transients. Or, set it just high enough to avoid pops and clicks
- **Drift**—Determines the likelihood of a slice changing after it has played
- **Playback**—Sets the playback level of the loop sequence
- **Low Cut and Hi Cut**—Adjust these to filter some of the bass or treble frequencies of the loop playback
- **Interval 1 and Interval 2**—Sets the pitch of some slices, the likelihood of which is determined by the Pitch parameter (from an octave down to an octave up)



TIP: To only use pitch and reverse effects on your loop, turn Shuffle down to 0% and set the Slices and SeqLen to the same value.

6. Want to change it up? While the loop is playing, press the switch to randomize its slice sequence.

7. Quickly double-press the Looper switch.

Playback/recording stops, and the LED lights white, indicating a loop is in memory.

8. While the loop is playing or stopped, press and hold the switch.

The recording is deleted, and the LED lights a dim white.

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Snapshots

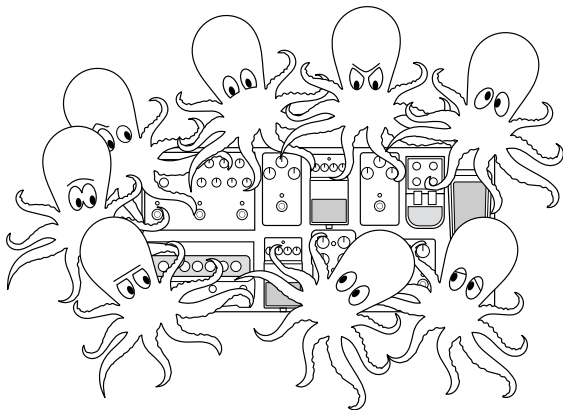
In Preset mode, the current preset's scribble strip displays a small camera icon. Its number indicates the current *snapshot*.



What are Snapshots?

Snapshots are presets within a preset.

Imagine you have eight pet octopuses, all slithering around your amp and pedalboard. Instead of tap-dancing on your pedals (and accidentally stepping on a tentacle), you shout, "Okay, gang—here's the verse... now!" Your octopuses switch some pedals on, switch other pedals off, and tweak all your pedals' knobs to make the best possible settings for your song's verse, all seamlessly with spillover delay and reverb trails. Then you shout, "Ready for the chorus... now!" Your octopuses instantly tweak everything for your song's chorus. That's the power of snapshots.



The only thing your octopuses/snapshots *can't* do is rearrange your pedalboard or swap out an effect for a different one (unless both effect blocks exist in the same preset).

Each preset can have up to 64 parameters assigned to Snapshots; hence, eight octopuses with eight tentacles each. The octopuses can remember four separate groups of on/off statuses and setting tweaks per preset; that is, HX Effects has four snapshots per preset.

Each of the four snapshots in HX Effects stores and recalls the state of certain elements in the current preset, including:

- **Block Bypass**—The bypass (on/off) state of all processing blocks (except Looper).



NOTE: The bypass state of blocks is automatically stored and recalled per snapshot. Optionally, you can exclude a block's bypass state from being affected by snapshots—see Snapshot Block Bypass On/Off section below.

- **Parameter Control**—The values of any parameters assigned to controllers (up to 64 per preset).
- **Command Center**—The values of any instant MIDI CC, Bank/Prog, MMC, and HX Preset/Snapshot/Looper messages, plus the state (dim or lit) of any CC Toggle or Ext Amp messages. Also, see "[Command Center](#)".
- **Tempo**—The current system tempo, if "[Global Settings > MIDI/Tempo](#)" > Tempo Select is set to "Per Snapshot." (By default, it's set to "Per Preset")

Depending on how you set them up, snapshots can act as **four variations of the same tone, four drastically different tones, or any combination thereof—all within the same preset**. A single preset's snapshots may accommodate all the various tones required for a song in many cases.



IMPORTANT! Snapshots make it easy for multiple blocks assigned to the same footswitch to end up in unexpected states. For example, if FS2 toggles between Delay (ON) and Reverb (OFF) blocks and a snapshot turns the Reverb on, FS2 will suddenly turn both blocks on and off together.

Snapshot Block Bypass On/Off

There may be situations where you *don't* want snapshots to control a particular block's bypass state; that is, you'd prefer to turn it on and off manually. For example, you may want manual control of a Boost block at any time without worrying whether a particular snapshot might turn it on or off.

1. When in Stomp footswitch mode, touch the footswitch assigned to the block you wish to exclude to select it, and press ACTION.
2. Press Snapshot Bypass to toggle On/Off state for this block.



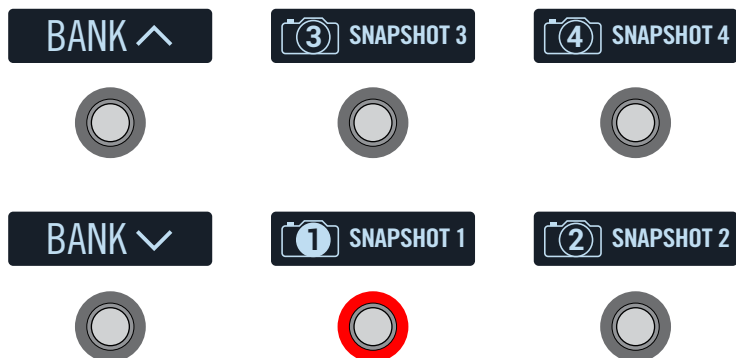
By default, all blocks are set to "On" to automatically control their bypass state via snapshots. Change to "Off" to disable snapshot control of the selected block's bypass state.



NOTE: The arrow at the right of this scribble strip indicates that there are more Action options on the next "page." (You may alternatively see this > indicator by itself on a scribble strip, depending on the type of block or menu selected.) Press the > button to advance to the next page for other features.

Using Snapshots

1. From Preset mode, press **BANK ^** and **BANK v** simultaneously to enter Snapshot footswitch mode.



2. Press one of the four snapshot switches to select a different snapshot.

NOTE: Optionally, you can configure these ["Global Settings > Switches"](#) options, which affect Snapshot switching:

- If you don't want Snapshot footswitches to disappear after selecting one, set Snpst Mode (page 2, knob 1) to "Latch[ing]."
- If you prefer to have the two left footswitches set for SNAPSHOT ^/v (instead of BANK ^/v), set Up/Dn Switch (page 2, knob 2) to "Snpst."
- If you prefer to invert two left footswitches' ^/v behaviors so that they are v/^, set Swap Up/Dn (page 2, knob 3) to "On."

3. Adjust the preset by doing one or more of the following:

- Turn one or more effects on or off by pressing stomp mode footswitches or pressing Footswitch 3 in the ["Signal Flow"](#) view. By default, snapshots automatically remember every block's bypass state, but you can change this behavior—see ["Snapshot Block Bypass On/Off" on page 31](#).
- To adjust a parameter AND have it automatically update per snapshot, press and turn the knob. This creates a "Snapshot controller assignment" for this parameter. Snapshots can remember the values of up to 64 effects parameters. The parameter's value bar displays small nodes at the current value, indicating the Snapshot controller's assigned to it:



SHORTCUT: Hold ACTION and press a parameter knob to quickly remove any controller assignment (including the Snapshots controller). The small value nodes disappear, indicating no controller is assigned to it.

NOTE: You can also manually assign the Snapshots controller (see ["Assigning a Controller"](#)). From the **Controller Assign** page, select the desired block and parameter and turn Knob 2 (Controller) to select "Snpst."

- On the ["Command Center"](#) page, adjust the values of any Instant ⚡ messages or press a footswitch assigned to CC Toggle, HX Preset/Snapshot/Looper message, or Ext Amp. Snapshots remember the values of any instant MIDI CC, Bank/Prog, MMC messages, and HX Looper messages, plus the state (dim or lit) of any CC Toggle and Ext Amp messages.

4. Switch back to the snapshot you started with.

The HX Effects device instantly and seamlessly returns to its previous state. Be sure to save to retain all your snapshot settings with the current preset.

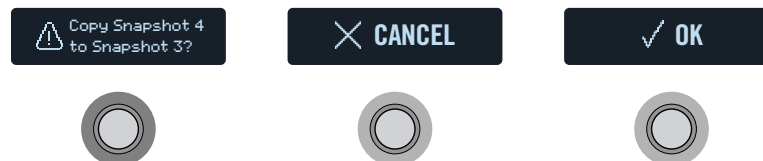
NOTE: If you've changed ["Global Settings > Preferences"](#) > Snpst Edits to "Discard," you must save the preset before selecting a different snapshot; otherwise, any edits will be discarded!

Copying/Pasting a Snapshot

Instead of creating a new snapshot from scratch, you may want to copy an existing one to another snapshot location and tweak just a few things.

1. From Snapshot mode, while touch-holding the footswitch for the snapshot you want to copy, briefly touch and release the footswitch for the snapshot you want to overwrite.

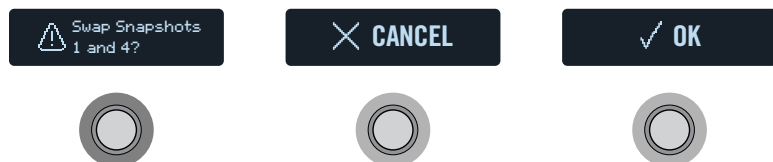
A dialog panel appears:



2. Press the OK switch.

Swapping Snapshots

1. Touch (but don't press) the two snapshot switches you want to swap until the following dialog appears:



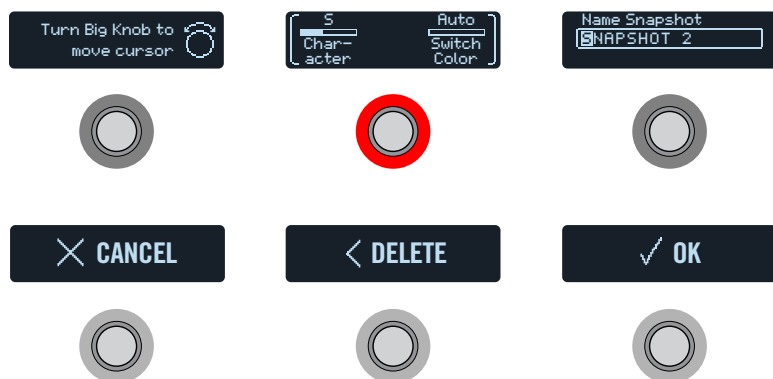
2. Press the OK switch.

Customizing a Snapshot Footswitch's Name and Color

Instead of having to remember the difference between "SNAPSHOT 1" and "SNAPSHOT 2," it's helpful to rename your snapshots something descriptive, such as "VERSE," "BIG SOLO," or "D. IGLOO," and/or give the switch LED a unique color.

1. From Snapshot mode, select the snapshot you want to rename and press ACTION.
2. Press the NAME SNAPSHOT switch.

The Name Snapshot screens appear:



Turn the Big Knob to move the cursor.

Turn Knob 1 (Character) to change the selected character.

 **SHORTCUT:** Press Knob 1 (Character) to cycle through uppercase, lowercase, 0, and [SPACE].

Press the DELETE switch to delete the selected character and shift all following characters to the left.

Delete all characters to remove the custom label, after which the footswitch label displays its normal assignment.

3. Turn Knob 3 (Switch Color) to select the desired color for the footswitch's LED ring (or turn it off).

Normally, you should leave this set to "Auto Color."

4. When finished, press the OK switch.

Saving Snapshots

Press SAVE twice to save the preset.

Saving a preset stores all of its four snapshots automatically.



NOTE: Selecting a preset recalls the Snapshot that was active when the preset was last saved.

Determining Snapshot Edit Behavior

Say you're on Snapshot 2 (VERSE), and you change a few things—switch a delay block on, switch a mod block off, tweak a (Snapshot Controller assigned) parameter, etc. If you switch to Snapshot 4 (CHORUS) and then back to Snapshot 2 for the second verse, should the HX Effects hardware recall those changes or return Snapshot 2 to its state when the preset was last saved? There's no right answer. Therefore, the HX Effects device lets you choose.

1. Press  and then the GLOBAL SETTINGS switch.
2. Turn the Big Knob to select Prefs, then touch the footswitch displaying "Snapst Edits" (FS2).
3. Turn the corresponding knob to set snapshot edit behavior.
 - **Recall**—Any snapshot edits are recalled when jumping from snapshot to snapshot and appear as you *last left them* (the default).
 - **Discrd** [Discard]—Any snapshot edits are discarded when jumping from snapshot to snapshot and appear as you *last saved* the preset. If you want to save changes made to a snapshot while Snapshot Edits is set to "Discard," **press SAVE twice before selecting another snapshot.**

Snapshot Reselect

This additional ["Global Settings > Preferences"](#) > Snap Reselect parameter adds even more flexibility to the Snapshot footswitches. For example, say you're playing along on Snapshot 2 (VERSE), then you press the Snapshot 4 (CHORUS) to load it. What should happen if you press the Snapshot 4 footswitch again? With Snapshot Reselect set to the default "Reload," Snapshot 4 is simply reloaded again. But when set to "Toggle," it will now load the previously loaded snapshot (Snapshot 2, in this example)—and successive presses of the Snapshot 4 footswitch then continues to toggle between Snapshot 2 and Snapshot 4.

Tips for Creative Snapshot Use

- The obvious use case for snapshots is designating them to specific sections of your song. For example, Snapshot 1 would be the Intro, Snapshot 2 would be Verse 1, Snapshot 3 might be the Chorus, and so on.
- Turn any Delay, Reverb, and/or FX Loops blocks' Trails parameter to "On" for seamless spillover between snapshots.
- "SNAPSHOT (X)" isn't very descriptive. Don't forget to name your snapshots—See ["Customizing a Snapshot Footswitch's Name and Color"](#)
- Worried that further tweaking might make your tone worse, not better? Snapshots are a great way to compare minor changes within a preset without having to take your hands off the guitar.
- Want to save your footswitches for other uses? Command Center MIDI type Instant ⚡ messages are automatically transmitted when a Snapshot is recalled.
- In a preset that includes a Looper block, open Command Center, and configure an Instant ⚡ command with the HX Looper - Play message assigned for Snapshot 3, and with the HX Looper - Stop message assigned for Snapshot 4. Now record a loop, and you can toggle between Snapshot 3 and 4 to play and stop your loop (along with any other actions you might additionally set to happen on these two snapshots).
- Set different keys in Harmony Delay blocks or intervals in Pitch blocks per snapshot.
- Having difficulty maintaining consistent volume throughout a song? Set one of the effects' Gain or Level parameter per snapshot.
- Remember to SAVE your current preset to retain any Snapshot settings you've created or edited before loading another preset!

Signal Flow

Although most users should expect to spend the vast majority of their time in Stomp and Preset modes, the Signal Flow menu is where you can optionally add or change blocks, reorder effects (independent of footswitch location), assign multiple effects to the same switch, and create parallel routings.

Signal Flow view also provides a great way to see all nine effects at once.

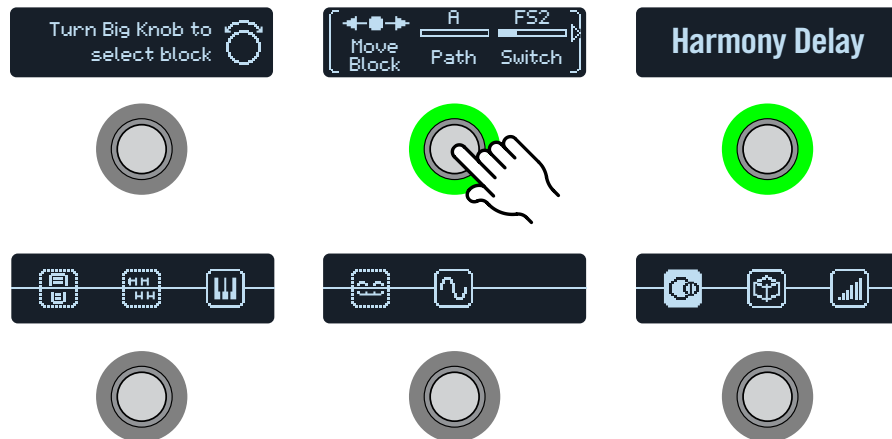
1. Press  to open the Menu.

2. Press the SIGNAL FLOW switch.

Up to nine simultaneous effects are represented by blocks spanning the bottom three screens. The currently selected block's model name appears at Footswitch 3.

To select a block, turn the Big Knob.

Touch FS2 for block moving, routing, and footswitch assignment.



Adding a Block in Signal Flow View

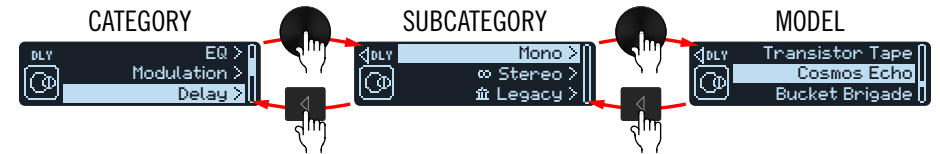
1. Turn the Big Knob to select an empty location in the path.

An empty block location appears as a small, solid square when selected:



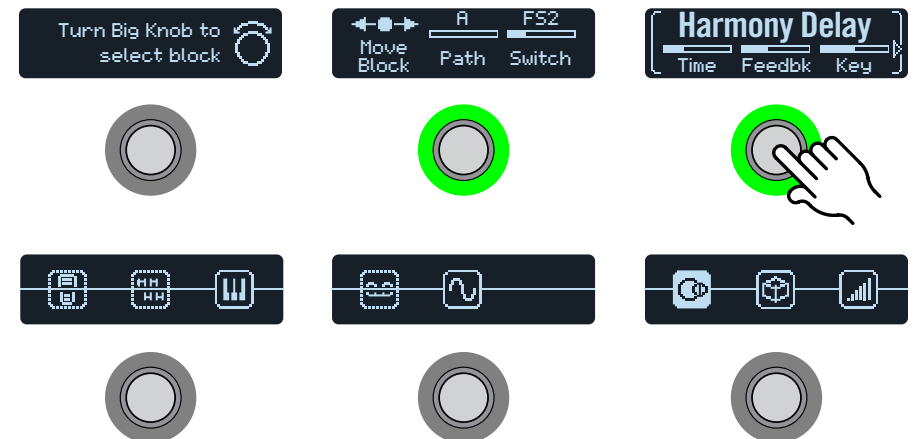
Note that you can move or clear existing blocks to make room for adding your new desired block type—see the following sections.

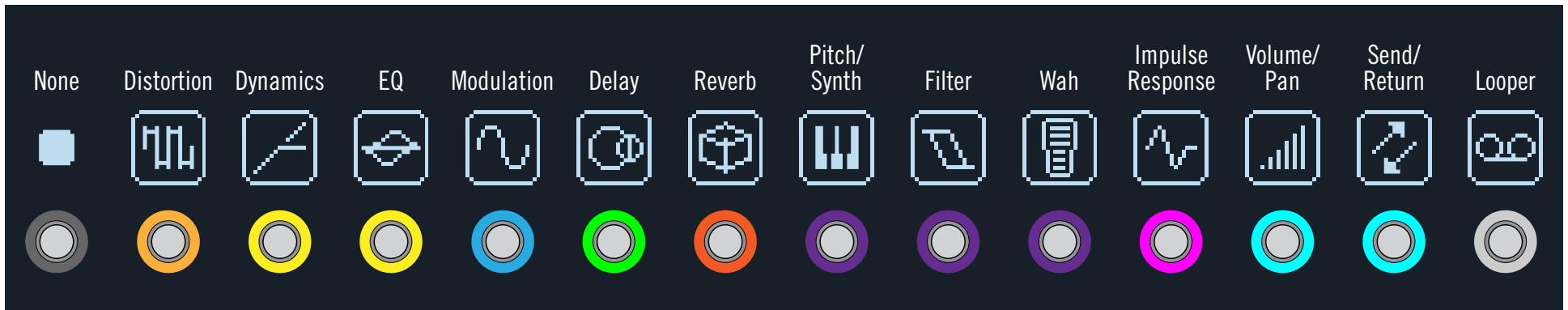
2. Press the Big Knob to open the model menu, which appears in the top right scribble strip. Choose your desired model.



TIP: You can also create your own Favorites—versions of effects models saved with your preferred settings. Once created, they'll appear in a "Favorites" category at the top of the list. See ["Adding a Block to Favorites"](#)

Touch FS3 to edit the selected block from within Signal Flow view.





Moving Blocks in Signal Flow View

1. Turn the Big Knob to select the block you want to move.
2. Touch Footswitch 2 and turn Knob 1 (Move Block) to move the selected block left or right.

Bypassing Blocks in Signal Flow View

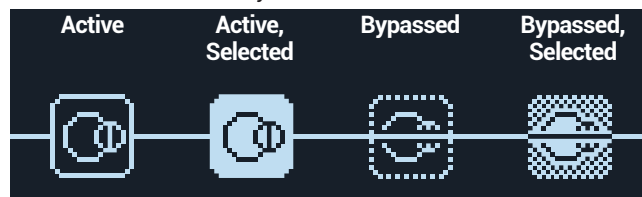
A Stomp mode footswitch is the most obvious method for turning an effect block on and off, but the HX Effects device can also engage or bypass a block automatically when moving an expression pedal. For example, if you add a Wah, Pitch Wham, or Poly Wham block via the Signal Flow view, you'll see that these block types are automatically assigned so that the block's Bypass is toggled via EXP 1 - moving EXP 1 forward past the heel down position enables the block, and returning EXP 1 to the heel position will bypass it again.

1. Turn the Big Knob to select the block you want to bypass.

You'll see the LED ring for FS2 and FS3 appear lit if the selected block is active or dimmed if bypassed.

2. Press FS2 or FS3 to bypass the selected block.*

An effects block's appearance changes depending on its state. For example, here are the various states of a Delay block:



When a parallel path is created, you'll see "Split" and "Mixer" blocks added (see [«Parallel Path Routing»](#)). The Mixer block cannot be bypassed. If the Split block is bypassed, it behaves exactly like Split > Y, with both Balance parameters set to Center.



***IMPORTANT!** Pressing FS2 or FS3 from the Signal Flow view bypasses only the selected block, even if other blocks share the same footswitch within Stomp mode. For example, if both a distortion (active) and delay (bypassed) block are assigned to FS1, pressing FS1 while in Stomp mode will toggle between distortion and delay. In such a case while in Stomp mode, repeatedly touch the multi-assigned FS1 to cycle through the distortion and delay parameters.

Editing Blocks in Signal Flow View

1. Turn the Big Knob to select the block you want to edit.
2. Touch Footswitch 3 and turn Knobs 1-3.

While adjusting parameters, the values briefly appear on the scribble strip:



If an effect or other item has more than three parameters, arrows appear to the right and/or left of the parameter names (see above).

3. Press ◀ or ▶ to access more parameters (if available).



TIP. Just like in Stomp mode, blocks can be copied, pasted, and cleared from the Signal Flow view. Press ACTION and then the desired function switch. See ["Copying/Pasting an Effect"](#)

Assigning Switches in Signal Flow View

IMPORTANT! The first six effects blocks added in the Signal Flow view are assigned to Stomp switches automatically. Don't worry about the order in which you add blocks, and in most cases, don't even bother attempting to reorder your Stomp switches from the Signal Flow view's Footswitch parameter. Instead, you can swap footswitch assignments by touching two Stomp switches. See ["Swapping Footswitches"](#).

However, Signal Flow view's Footswitch parameter overwrites any automatic assignment and lets you assign multiple effects to the same switch, and even allows for toggling between two or more effects.

1. Turn the Big Knob to select the block you want to assign to a footswitch.
2. Touch Footswitch 2 and turn Knob 3 (Switch) to select the desired footswitch or expression pedal:

None Removes the switch assignment.

FS1-FS6 Stepping on the Stomp mode footswitch turns the block on and off.

Selecting FS1-6 displays Knob 1 (Switch Type) on the second page. Press **▷** and turn Knob 1 (Switch Type) to select **"Latch[ing]"** or **"Moment[ary]"**. When set to Moment, the block is bypassed (or enabled, if already bypassed) for as long as you hold the switch. When set to Latch, the block is bypassed (or enabled, if already bypassed) every time you press the switch.

EXP 1, EXP 2 Configures the expression pedal's movement to automatically enable (or bypass) the block, offering the following options:

Selecting EXP 1 or 2 displays Knob 1 (Position) and Knob 2 (Wait) on the second page. Position determines at which point within the expression pedal's travel the block is enabled or bypassed. 0% is heel down; 99% is toe down. Wait determines how long the HX Effects device waits before bypassing the block. For example, you wouldn't want the wah to turn off every time you touched the heel down position in your big funk wah solo. Knob 1 on page 3 (Behavior) determines how bypassing is handled. When set to "Toggle," bringing the pedal past the Knob 2-Position value will toggle the block's bypass state from what it is currently (i.e. enable if currently bypassed and vice versa). "Toe Down" and "Heel Down" always bypass the block at the designated position, regardless of the block's starting state. (With "Heel Down" selected, the block will always enable when you cross the page 2 - Knob 2 - Position value and bypass when you return below it.)



TIP: By default, bypass toggling the effect via EXP 1 or EXP 2 is configured for "heel down = off" behavior. To reverse bypass behavior, press FS3. In such a case, the block will be bypassed when moving the expression pedal past the configured Position location. Since it is possible to configure the bypass of multiple blocks to be assigned to an expression pedal, you can set each block's Position value differently, thereby allowing the pedal to turn some blocks on and others off at different positions in the pedal's travel.

You can change additional bypass behaviors for the selected block within the Signal Flow view by touching FS2 and accessing both pages' parameters.



NOTE: Footswitch type (momentary or latching) is determined per footswitch, not per assignment.

3. If desired, press **▷** and turn Knob 3 (MIDI In) to assign an incoming MIDI CC message to turn the block on and off.

Incoming CC values 0-63 turn the block off; values 64-127 turn the block on. Note that some MIDI CCs are reserved for global functions and cannot be selected.



NOTE: When assigning a parameter to a Footswitch or MIDI, Snapshots are also assigned by default. However, there's an additional **Snapshot Control** option within the Controller Assign parameters—set this to "Off" to bypass this behavior.

Parallel Path Routing

The vast majority of guitar tones are created with serial routing; that is, one stompbox or effect after another. For example, the Signal Flow view for most of the HX Effects device's presets looks something like this:



For more sophisticated tones, a parallel (two stereo paths) signal flow can be created. This lets you split the signal into two stereo paths, process each separately, and mix the two back together.

A couple advantages of parallel routing:

- If a reverb follows a delay on a serial path, the delay's echoes will have reverb applied. Likewise, if a delay follows a reverb on a serial path, the reverb's tail will have distinct echoes. Conversely, if a reverb and delay are on separate parallel paths, they won't affect one another and can sometimes result in cleaner, more defined notes (if desired).
- If your effect block types don't have Mix or Blend controls, adding them to a parallel path allows the dry guitar or bass signal to pass through unaffected and be blended with the effected signal.

Touch Footswitch 2 and turn Knob 2 (Path) to move the selected block down to Path B.

The block is dropped to lower path B, a Split block is created directly to the left of it, and a Mixer block is created after the last effect:



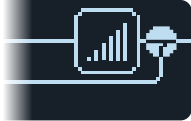
To remove path B, move all blocks on path B (lower) back up to path A (upper).

Setting Path B's Output

There may be situations where you want parallel Path B to be sent to a completely different set of outputs.

1. In the Signal Flow view, turn the Big Knob to select the point where the two paths merge.

The Mixer block appears only when selected:



2. Touch Footswitch 2 and turn Knob 2 (Path B Output) to select "Send12."

The Mixer block splits into two, indicating Path A is being sent from the HX Effects hardware's Main L/R outputs and Path B is being sent from its Send 1/2 outputs:



Configuring Split Block Options

It is possible to customize a Split block's options to determine how your signal is fed into Paths A and B.

TIP: Remember that both Paths A and B are stereo signal paths. Therefore, the same routing and block principles still apply to each Path—see ["Block Order and Stereo Imaging"](#) on page 40.

1. In the Signal Flow view, turn the Big Knob to select the Split block, at the point where the two paths split.

The Split block appears only when selected.



By default, the **Split Y** type block is selected, as displayed on FS3. Adjust the amount of signal fed to each path using the **BalanceA** and **BalanceB** parameters. Optionally, you can choose a different Split block type with different routing options.

2. Touch Footswitch 2 and press the Big Knob.

The FS3 scribble strip now shows the menu where you can choose between the four available Split types: **Y**, **A/B**, **Crossover**, and **Dynamic**—described in the following tables.

Split > Y Settings

By default, a Split > Y appears any time parallel Path B is created.

Knob	Parameter	Description
1	Balance A	Adjusts the stereo balance (pan) of the signal sent to Path A (upper).
2	Balance B	Adjusts the stereo balance (pan) of the signal sent to Path B (lower).

Split > A/B Settings

The signal can be sent in different amounts to Paths A (upper) and B (lower).

Knob	Parameter	Description
1	RoutTo	Determines the amount of the signal sent to Path A vs. Path B. Press the knob to set to Even Split .

Split > Crossover Settings

Treble frequencies are sent to Path A, and bass frequencies to Path B.

Knob	Parameter	Description
1	Freq	Any signal above this frequency is sent to Path A (upper path); any signal below this frequency is sent to Path B (lower path).
2	Reverse	When on, reverses the path assignments (any signal above the crossover frequency is sent to Path B, any signal below the crossover frequency is sent to Path A).

Split > Dynamic Settings

Page	Knob	Parameter	Description
1	1	Threshold	Any signal below the Threshold volume level is routed to Path A; any signal above the Threshold is routed to Path B.
	2	Attack	Determines how fast the signal is routed to Path B once reaching the Threshold.
	3	Decay	Determines how fast the signal returns to Path A once falling below the Threshold.
2	1	Reverse	When on, reverses the path assignments (any signal above the Threshold value is sent to Path B, any signal below the Threshold value is sent to Path A).



NOTE: Much like any Effects type block, a Split block can be bypassed and have a bypass assignment created for it. When bypassed, regardless of the Split type in use, the block sends both left and right signals to both paths equally.

Mixer Settings

Page	Knob	Parameter	Description
1	1	A Level	Adjusts the output level of Path A (upper path).
	2	A Pan	Adjusts the left/right stereo balance of Path A.
	3	B Level	Adjusts the output level of Path B (lower path).
2	1	B Pan	Adjusts the left/right stereo balance of Path B.
	2	B Polar	Inverts the polarity of Path B. Typically, this should be set to "Normal."
	3	Level	Adjusts the overall output level of the Mixer block.

Dynamic DSP

Like nearly all modern audio processors, the HX Effects engine is based on DSP (Digital Signal Processing). Some models require more DSP horsepower than others, so it makes logical sense that fewer of those models could exist within a preset. To work around this, some processors restrict you to one reverb, one delay, one pitch shifter, etc. With HX Effects hardware, we feel it's important to let you keep adding whatever you want to your Signal Flow, even if you may eventually run out of DSP.

That said, there are some rules governing the number of certain types of blocks you can add to a preset:

IR (Impulse Response) blocks Up to four 1024-point Single IR blocks (two per path max.), or two 2048-point Single IR blocks (one per path max.), or two Dual IR blocks, (one per path max.)

Polyphonic ("Poly") and high DSP type effects blocks One per path max. These effects include the Feedbacker (Dynamics), Poly Sustain (Delay), Poly Detune (Modulation), and Poly Pitch, Wham, Capo and 12 String (Pitch/Synth).

Looper block One

As you approach the max. DSP limit for the current preset, the HX Effects hardware's "Dynamic DSP" functionality automatically hides effects models from the menus that would result in exceeding the DSP limit. To follow are some tips to get the most out of your device.

Tips to Optimize DSP

- Some block types use much more DSP than others, such as Reverbs, the Dynamics - Feedbacker, and models that utilize polyphonic pitch shifting (Delay - Poly Sustain, Modulation - Poly Detune, and Pitch - Poly Pitch, Poly Wham, Poly Capo, 12-String) EQ, Dynamics, Volume/Pan, and Send/Return blocks use relatively little.
- Some models may use more DSP than others in the same category, therefore, you may find that choosing a similar type of effect within a category can reduce the DSP load, thus allowing you to utilize more blocks.
- The stereo version of an effects block will use roughly twice as much DSP as a mono version of the same block. Therefore, there is no sense to adding a stereo block before an IR or other mono type block (since the signal will be summed to mono anyway - see ["Block Order and Stereo Imaging"](#)).
- Some model categories have "Simple" blocks, which utilize less DSP than others.
- Instead of toggling between two of the same effects blocks (with different settings), use controllers or snapshots to instantly adjust parameters within a single block.

Block Order and Stereo Imaging

The HX Effects device signal path is stereo, carrying two channels of audio. When the device is fed a stereo input source (i.e. - when discrete left & right signals are connected into the device's Left and Right input jacks), the source audio is processed discretely in stereo, wherever stereo blocks are used within the path. Whenever a mono block is added within a path, both audio channels are combined and sent out of the block as mono.

Most effects models have both **Mono** and **Stereo** versions. The stereo imaging is highly dependent on the type of blocks you add and in what order.

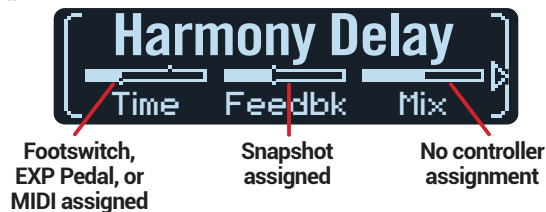
Legacy effects models vary in behavior, just as on the classic Line 6 effects from where these models originated.

- Legacy Distortion, Dynamics & Pitch/Synth effects are mono.
- Legacy Modulation & Delay effects vary in that some are mono, some stereo, and some mono-in/stereo-out, where adjusting the effect's Mix parameter can narrow the stereo image that is fed into them. It is best to experiment with these effects and tweak the Mix parameter to achieve the desired stereo output results.
- Legacy Filter and Reverb effects are stereo.

Controller Assign

The HX Effects device provides a wide variety of tools for controlling your tone during a performance. The most obvious controller would be a connected expression pedal (often assigned to Wah, Pitch Wham, or Volume). But, you may also assign footswitches to toggle between two values of a given parameter or parameters, control a parameter from an external MIDI device, or even have parameters instantly change when selecting different snapshots within a preset.

If a controller has been assigned to a parameter, small white nodes appear above and below its value bar:

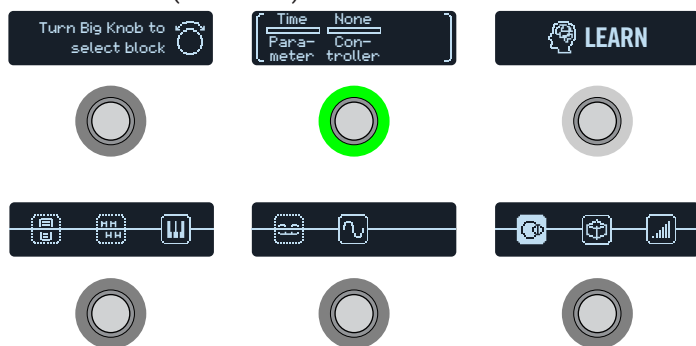


NOTE: Adding a Wah, Pitch Wham, or Poly Wham block via the Signal Flow view automatically assigns it to be controlled and bypassed by EXP 1. Adding a Volume Pedal or Pan block via the Signal Flow view automatically assigns it to be controlled by EXP 2. To change the bypass footswitch assignment for any block, see [page 37](#).

Assigning a Controller

1. From Stomp mode, press and hold the knob for the parameter you wish to control.

The HX Effects device jumps to the Controller Assign page and displays your parameter at Knob 1 (Parameter).



NOTE: To manually access the Controller Assign menu, press and then the **CONTROLLER ASSIGN** switch.

2. Press the **LEARN** switch.

The button's brain icon glows blue, and Footswitch 1 reads, "Press switch, move EXP, or send MIDI..."

3. **Step on a footswitch, move a connected expression pedal, send a MIDI CC message from your keyboard, etc.**

The controller name appears above Knob 2 (Controller).

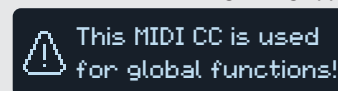
NOTE: To manually select a controller, turn Knob 2 (Controller).

- None** Removes the controller assignment.
- EXP 1 or 2** Expression pedals are the most common type of controller. Used to control Volume, Wah, Pitch Wham, etc.
- FS1-FS6** Stepping on a Stomp mode footswitch can toggle between a parameter's min and max values.

Selecting FS1-FS6 will display Knob 3 (Switch Type). **Turn Knob 3 to select "Moment[ary]" or "Latch[ing]."** Momentary means the value will change for as long as you hold the switch. Latching toggles between Min and Max values every time you press the switch.
- MIDI CC** Selecting "MIDI CC" will display Knob 3 (CC#). **Turn Knob 3 to select the desired MIDI CC number.**
- Snpsht** Although all controller-assigned parameters are updated per snapshot, an additional "Snapshots" controller is available when other controllers are already used.

NOTE: Footswitch type (momentary or latching) is determined per footswitch, not per assignment.

NOTE: The HX Effects hardware has reserved specific MIDI CC messages for global functions; these CCs cannot be used as controllers. If you attempt to learn a CC message reserved for global functions, the following dialog appears:



See ["MIDI"](#) for more information.

4. If desired, press and turn Knob 1 (Min Value) and Knob 2 (Max Value) to set the range you wish to control.

TIP: To reverse controller behavior, swap the min and max values.

5. Press to return to the Home screen.



SHORTCUT: To assign a parameter to the Snapshots controller, it's even easier—just push and turn the parameter's knob. The value bar displays small value pixels, indicating it's now assigned to a controller.



SHORTCUT: Hold ACTION and press a parameter knob to quickly remove any controller assignment (including the Snapshots controller). The value bar's controller nodes disappear, indicating no controller is assigned to it.

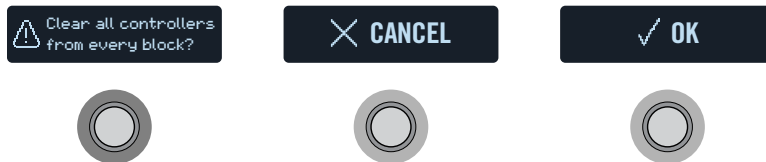
Clearing a Block's Controller Assignment(s)

1. From the Controller Assign screen, select the block whose controller assignments you want to clear and press ACTION.
2. Press the CLEAR CONTROLLERS switch.

Clearing All Controller Assignments

1. From the Controller Assign screen, press ACTION.
2. Press the CLEAR ALL CONTROLLERS switch.

The following dialog appears:



3. Press the OK switch.



IMPORTANT! Clearing All Controller Assignments removes *all* existing controller assignments in the preset, including Snapshot controller assignments and Wah, Pitch Wham, Poly Wham, Volume, and Pan assignments from EXP 1 and EXP 2. Use this function with caution!

Tips for Creative Controller Assignment

- If you assign a footswitch to more than one controller or other item, the default "MULTIPLE (X)" scribble strip label isn't really very descriptive. Don't forget to custom label it - See ["Customizing a Footswitch"](#)
- By default, a parameter's Min and Max values will be pretty extreme. It pays to play pretty conservatively here, as subtle parameter adjustments go a long way.
- To smoothly blend between the tone on parallel paths A and B, select a Split > A/B block and assign the Route To parameter to an expression pedal. By default, a heel-down position means the signal passes fully through Path A. Moving the pedal toward the toe-down position will gradually crossfade into Path B. Alternatively, assign a footswitch to control the Route To parameter for instantly switching back and forth.
- If you're looking for an ultra-clean boost, instead of adding a Volume/Pan > Gain block, try assigning a footswitch to increase the Level parameter of a Merge > Mixer or Output block.
- If you have a favorite delay or reverb pedal, use an FX Loop block to insert it into your tone. Assign EXP 1 or 2 to control the block's Mix parameter, which will smoothly blend the pedal into your tone.
- For extreme psychedelic dub delay squeals, assign a footswitch to both increase a Delay's feedback, and decrease its time.
- Assign a footswitch to toggle between two Delay > Time parameter values, such as 1/4 and 1/8 dotted.
- Assign IR Select to a footswitch. Set the two IRs as min and max values. Now you can instantly toggle between them.

Command Center

The HX Effects device also just so happens to be a world-class master remote control for your entire pedalboard or live rig. Each of its Stomp mode footswitches and expression pedals can be used to send a variety of messages:

- Transmit MIDI or External Amp commands to your guitar amps, pedals, synths, or even other modelers.
- Send QWERTY Hotkey messages, emulating a computer keyboard, to practically any software to control it from your device (see ["QWERTY Hotkey Commands" on page 45](#)).
- Configure footswitches to expand your device's Stomp mode with additional control functions (see ["HX Preset, Snapshot, and Looper Commands" on page 45](#)).
- Also, up to six "Instant" ⚡ commands can be transmitted automatically when an HX Effects preset is recalled, for starting your DAW, triggering a MIDI-controlled lighting system, or switching presets on external gear.

NOTE: Instant commands offer a **Wait Time** parameter, letting you delay certain events by up to 1000ms. This is to help Helix control older MIDI gear that may behave incorrectly when receiving multiple commands very close together. Unless you specifically need to delay a message, leave this set to "0 ms."

All Command Center assignments are stored per preset, but they can be copied and pasted to other presets. See ["Copying and Pasting a Command"](#)

NOTE: The Value parameters of any instant MIDI CC, Bank/Prog, MMC, and HX Looper messages, plus the state (dim or lit) of any CC Toggle and Ext Amp messages are automatically recalled when selecting a snapshot.

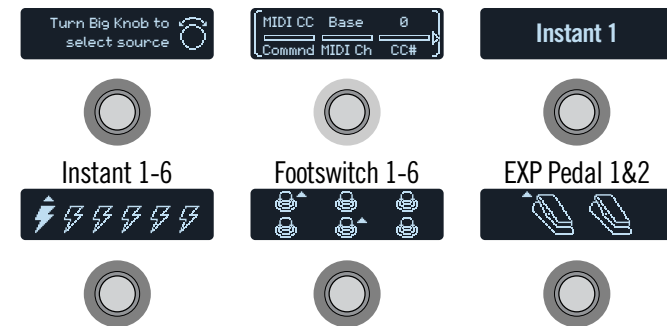
NOTE: All MIDI-based Command Center messages are transmitted via MIDI Out and USB simultaneously. See ["Global Settings > MIDI/Tempo"](#) for MIDI configuration options.

Assigning a Command

1. Press **≡** to open the Menu.

2. Press the **COMMAND CENTER** switch.

The Command Center screens appear, with the command sources in the HX Effects hardware represented by 14 icons along the bottom three screens:



3. Turn the Big Knob to select the command source (Instant 1-6, Footswitch 1-6, or EXP Pedal 1-2).

The source name appears at Footswitch 3.

4. Turn Knob 1 (Command) to select the type of command you wish to transmit.

Not all command sources can send the same types of commands.

Select "None" to remove any existing command assignment.

5. Press **◀** or **▶** and turn Knobs 1-3 to adjust the command's settings, which are determined by the type of command selected:

MIDI CC (Continuous Controller)			
Page	Knob	Parameter	Description
1	2	MIDI Ch	Sets the CC message's MIDI channel (1-16). When set to "Base," the HX Effects device follows the Global MIDI channel, which is set from the "Global Settings > MIDI/Tempo" page.
	3	CC #	Sets the CC number (0-127).
2	1	Value, Min Val[ue]	Sets the CC number's value (0-127). For EXP Pedal 1 and 2, sets the minimum CC value controlled by the pedal.
	2	Max Val[ue]	For EXP Pedal 1 and 2, sets the maximum CC value controlled by the pedal.
CC Togl [Toggle]			
Page	Knob	Parameter	Description
1	2	MIDI Ch	Sets the CC messages' MIDI channel (1-16). When set to "Base," the HX Effects device follows the Global MIDI channel, which is set from the "Global Settings > MIDI/Tempo" page.
	3	CC #	Sets the CC number (0-127).

CC Togl [Toggle]			
Page	Knob	Parameter	Description
2	1	DimVal[ue]	Sets the CC number's value (0-127) when the footswitch ring is dim.
	2	LitVal[ue]	Sets the CC number's value (0-127) when the footswitch ring is lit.
	3	Latch	Sets the footswitch's behavior as either "Latching (the default) or "Momentary."

 **NOTE:** For CC Toggle commands, one of two values is automatically transmitted upon preset recall, determined by the footswitch's state (dim or lit) when the preset was saved. Subsequent presses of the footswitch toggle between the two states' CC values—Dim Value and Lit Value.

Bank/PC (Program Change)			
Page	Knob	Parameter	Description
1	2	MIDI Ch	Sets the Bank/Program message's MIDI channel (1-16). When set to "Base," the HX Effects device follows the Global MIDI channel, which is set from the "Global Settings > MIDI/Tempo" page.
	3	CC00	Sets the CC#00 (Bank MSB) value. Select "Off" if the receiving device shouldn't respond to Bank MSB.
2	1	CC32	Sets the CC#32 (Bank LSB) value. Select "Off" if the receiving device shouldn't respond to Bank LSB.
	2	Prog[ram]	Sets the Program Change (PC) value. Select "Off" if you only want to send a Bank MSB and/or Bank LSB message.

Note On			
Page	Knob	Parameter	Description
1	2	MIDI Ch	Sets the note's MIDI channel (1-16). When set to "Base," the HX Effects device follows the Global MIDI channel, which is set from the "Global Settings > MIDI/Tempo" page.
	3	Note	Sets the MIDI note value (C-1 ~ G9). Middle C is C3.
	1	Velcity	Sets the MIDI note's velocity (0-127).
2	2	NoteOff	Determines whether the MIDI note sustains until pressing the switch again (Latching) or sustains only while the switch is held (Momentary).

MMC (MIDI Machine Control)		
Knob	Parameter	Description
2	Message	Determines the message type.
Ext Amp		
Knob	Parameter	Description
2	Select	Determines the Ext Amp connection [1 (Tip-to-sleeve), 2 (Ring-to-sleeve), or both] for switching an external amp's channel, reverb, or other functions. When assigned to an Instant command, there is also the option to choose "None," thereby allowing you to send a different (or None) message per snapshot.
3	Type	When assigned to a footswitch, determines whether the footswitch behavior is Latching (default) or Momentary.

 ***IMPORTANT!** Connect EXT AMP 1/2 only to amplifiers that utilize "short-to-sleeve" footswitch inputs. Connecting to any other sort of input could cause permanent damage to both your amp and the HX Effects device! If you're not sure if your amp has short-to-sleeve inputs, contact the manufacturer.

 **IMPORTANT!** Your ability to control external amp channel and/or reverb switching with the HX Effects device has been tested with many popular amps and heads. Unfortunately, this does not guarantee compatibility with all products. Note that, depending on the circuitry of the channel switching jack in the guitar amp used, the EXT Amp function may not operate as expected.

QWERTY Hotkey Commands

QWERTY hotkeys (computer keyboard shortcuts with or without modifiers, such as Shift, Control, Option/Alt, and Command) can also be sent to your Mac, PC, or iOS device via USB, allowing you to control virtually any DAWs, YouTube, Spotify, looping software, DJ software, media players, lighting software, and more!

QWERTY command assignments can be made to any footswitch, or to any Instant ⚡ command, allowing them to be sent automatically from any preset or snapshot recall. These commands can be assigned on HX Effects using the preceding steps within this chapter (but you'll likely find it faster and easier using the HX Edit app). Just like a standard computer keyboard, hotkeys won't work unless the software or app you want to control has focus.

QWERTY Hotkeys			
Page	Knob	Parameter	Description
1	2	Mod 1	Set up to three key modifier(s) to be sent with the key-stroke: Choose Shift, Alt, Ctrl, or Mac/PC (corresponds to the "Windows" key on Windows OS systems, and the "Command" key on macOS systems.
	3	Mod 2	
	1	Mod 3	Choose None for knob 2~4 for no added modifier.
2	2	Keystk	Sets the alpha, numeric, or other computer keyboard keystroke value to be sent.
	3	Type	Sets the footswitch type as either Momentary or Latching.

 **TIP:** To get you started, we've provided several factory presets pre-configured with hotkey commands to remotely control numerous popular DAW, multi-media, and productivity computer apps!

HX Preset, Snapshot, and Looper Commands

These "HX" commands allow you to configure Stomp mode switches for even greater control of your device's internal functions. HX commands can be assigned on HX Effects using the preceding steps (see ["Assigning a Command"](#)). All assignments are saved per preset.



NOTE: Although Stomp mode switches can be assigned to multiple functions, to avoid unintended behavior, we strongly recommend the HX Preset, HX Snapshot, and HX Looper commands be assigned only to empty footswitches.

HX Preset

The HX Preset command allows you to configure a Stomp mode footswitch to immediately "jump to" any other preset within the current setlist.

Preset (HX Preset)		
Knob	Parameter	Description
2	Preset	Sets the preset to be selected by the command: Next, Previous, or a preset's number (01A~32D).

HX Snapshot

The HX Snapshot command allows you to configure a Stomp mode footswitch to recall any snapshot within the current preset immediately.

1. From the Command Center page, select a footswitch and turn Knob 1 (Command) to "Snpsht."
2. Turn Knob 2 (Behavior) to either "Press/Release" or "Press/Hold."
3. Turn the other two knobs to configure which snapshots are recalled per the "Press" and the "Release" (or "Hold") of the footswitch, respectively.

Snpsht (HX Snapshot)			
Page	Knob	Parameter	Description
1	2	Type	Sets the behavior for the footswitch. This allows you to recall the desired snapshot index number, or Next or Previous snapshot, <i>independently</i> per each Press and Release , or each Press and Hold of the footswitch. (Note that the assigned, Stomp mode footswitch's label and LED will only indicate the command assigned to the Press function.)
	3	Press	Sets the snapshot to be recalled on the Press action of the footswitch.
2	1	Release/Hold	Sets the snapshot to be recalled on the Release (or Hold, depending on your page 1, Knob 2 selection) of the footswitch.

HX Looper

For a preset that includes a Looper block, this command allows you to configure an Instant ⚡ or Stomp mode footswitch to trigger any function of the Looper.

 **NOTE:** A 6 Switch or 1 Switch Looper block must be present in your preset for HX Looper commands to function.

HX Looper - Instant Command Assignment:

1. In the Command Center window, select an Instant ⚡ command controller and turn Knob 1 (Command) to “Looper.”
2. Turn Knob 2 (Function) to choose the specific Looper function you wish to trigger.

The selected **Function** action is automatically stored and recalled per snapshot. This allows you to record a loop and, for example, configure an Instant command to automatically “Play” the Loop when loading one Snapshot and “Stop” when loading a different Snapshot.

Looper (HX Looper) - Instant ⚡ Command		
Knob	Parameter	Description
2	Function	Sets the Looper action to be triggered: Play, Stop, Play/Stop, Rec, Overdub, Record/Overdub, Play Once, Reverse, Forward, Reverse/Forward, Half Speed, Full Speed, Toggle Speed, or Undo.

HX Looper - Footswitch Assignment:

1. In the Command Center window, select a footswitch and turn Knob 1 (Command) to “Looper.”
2. Turn Knob 2 (Behavior) to either “Prs/Rel” or “Prs/Hld.”
3. Turn the other knobs to configure which Looper commands are sent per the “Press” and the “Release” (or “Hold”) of the footswitch, respectively.

The selected **Behavior**, **Press**, and **Release/Hold** actions are automatically stored and recalled per snapshot.

Looper (HX Looper) - Footswitch			
Page	Knob	Parameter	Description
1	2	Type	Sets the behavior for the footswitch. This allows you to send up to two different Looper commands, <i>independently</i> per each Press and Release or each Press and Hold of the footswitch. (Note that the assigned, Stomp mode footswitch's label and LED will only indicate the command assigned to the Press function.)
	3	Press	Sets the Looper command to be sent on the Press action of the footswitch.
2	1	Release/Hold	Sets the Looper command to be sent on the Release (or Hold, depending on your page 1, Knob 2 selection) of the footswitch.

Note that the assigned, Stomp mode footswitch's label and LED will only indicate the command assigned to the **Press** function.

Copying and Pasting a Command

1. Turn the Big Knob to select the source containing the command you wish to copy and press ACTION.
2. Press the COPY COMMAND switch.
3. Select the destination to which you want to paste the command—even in a different preset—and press ACTION.
4. Press the PASTE COMMAND switch.

Copying and Pasting All Commands

Setting up the same or similar set of Command Center commands across multiple pre-sets can quickly become tiresome. Fortunately, the HX Effects device lets you quickly copy and paste all commands to another preset.

1. From the Command Center page, press **ACTION**.
2. Press the **COPY ALL COMMANDS** switch.
3. Select the preset to which you want to paste the commands and press **ACTION**.
4. Press the **PASTE ALL COMMANDS** switch.

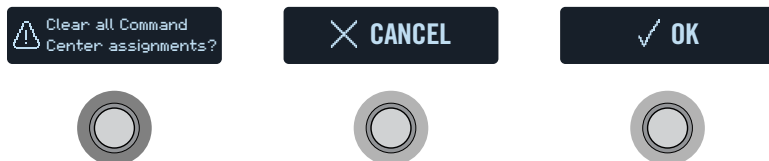
Clearing a Command

1. Select the source containing the command you want to clear and press **ACTION**.
2. Press the **CLEAR COMMAND** switch.

Clearing All Commands

1. From the Command Center page, press **ACTION**.
2. Press the **CLEAR ALL COMMANDS** switch.

The following dialog appears:



3. Press the **OK** switch.



TIP: Be sure to **SAVE** your preset after adding, editing, or clearing Command Center commands to retain your changes.



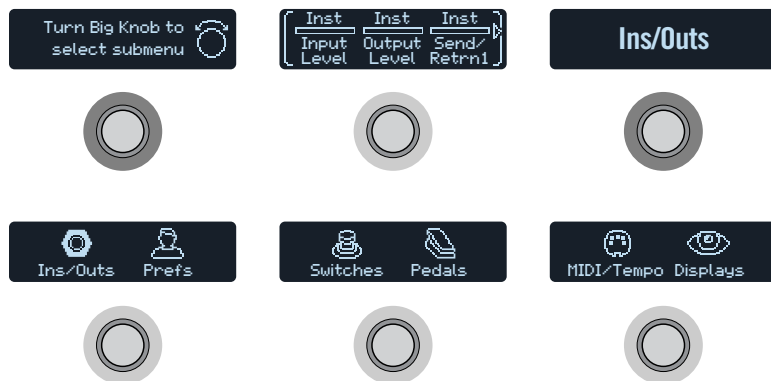
TIP: You can customize the footswitch's label and LED color if desired—see [“Customizing a Footswitch” on page 15](#)

Global Settings

The Global Settings menu contains additional parameters that apply to all presets, such as input and output levels, footswitch configurations, etc.

1. Press  to open the Menu.
2. Press the GLOBAL SETTINGS switch.

The Global Settings screens appear:



3. Turn the Big Knob to select one of the six submenus.

If necessary, press  to view more parameters.

 **NOTE:** See the next page for a description of all Global Settings.

Setting Proper Levels

The various inputs and outputs should be set to match that of your instrument, amp, and other pedals to ensure an ideal tone, low noise, and no ugly clipping distortion (unless, of course, you want ugly clipping distortion. Who are we to judge?). Just know that if you run to the Internet, screaming, "HX Effects sounds bad!" the first thing people will ask is if you read this section of the manual. And if you didn't, they'll promptly revoke your Guitar Gear Guru status.

1. From the Global Settings menu, turn the Big Knob to select the Ins/Outs submenu.

2. Using Knobs 1-3, set levels according to the following table:

Jack	What are you connecting?	Then do this:
INPUT L/MONO, RIGHT	Guitar or bass with passive pickups	Set Input Level to "Inst"
	Guitar or bass with really loud or active pickups	Set Input Level to either "Inst" or "Line" (trust your ears)
	Keyboard, synth, or drum machine	Set Input Level to "Line"
OUTPUT L/MONO, RIGHT	To the Guitar In on a guitar or bass amp	Set Output Level to "Inst"
	To the guitar input(s) of a stompbox or other multieffect	
	To the instrument-level effects return of a guitar or bass amp for post effects or 4-cable method	Set Output Level to "Line"
	To the line-level effects return of a guitar or bass amp for post effects or 4-cable method configuration	
	To the inputs of a studio rack effect or mixer	

Resetting All Global Settings

Resetting the HX Effects device's Global Settings returns them to factory default. Performing this reset does not affect any presets you may have created.

1. From the Global Settings menu, press ACTION.
2. Press the FACTORY SETTINGS switch.

The following dialog appears:



3. Press the OK switch.

Global Settings > Ins/Outs

Page	Knob	Parameter	Description
1	1	Input Level	Choose “Inst[rument]” when connecting a guitar, bass, or stompboxes to the HX Effects device’s input jack(s); choose “Line” when connecting synths, drum machines, the sends from mixers, or other line-level sources. If your guitar or bass has really loud active pickups, try both “Inst[rument]” and “Line” and trust your ears.
	2	Output Level	Choose “Inst[rument]” when connecting the 1/4" outputs to stompboxes or the front of guitar amps; choose “Line” when connecting to mixers or standalone recorders. When using a single amp or mixer channel, connect only the LEFT/MONO 1/4" jack.
	3	Send/Retrn1	Choose “Inst[rument]” when using a Send/Return pair as an FX loop for stompboxes; choose “Line” when using a Send/Return pair as an FX loop for line-level rack processors.
2	1	Send/Retrn2	

Global Settings > Preferences

Page	Knob	Parameter	Description
1	1	Bypass Type	When pressing MODE and TAP together, the HX Effects device bypasses all effects. There are two types of All Bypass in the HX Effects hardware—Choose “Analog” if you want your guitar’s signal routed directly from the HX Effects device’s inputs to its outputs with no processing or A/D/A conversion (also called “true bypass”). Choose “DSP” if you want delay echoes and reverb tails to decay naturally when the HX Effects device is bypassed.
	2	Snpsht Edits	Determines whether or not any edits made to a snapshot (block on/off, parameter control, Command Center, tempo) are remembered when returning to that snapshot. When set to “Recall,” any snapshot edits are recalled when jumping from snapshot to snapshot and appear as you last left them. When set to “Discrd,” any snapshot edits are discarded when jumping from snapshot to snapshot and appear as the preset was last saved. If you want to save changes made to a snapshot while Snapshot Edits is set to “Discrd,” press SAVE twice before selecting another.
	3	Tempo Pitch	Determines how delay repeats behave when repeatedly pressing TAP to change the Tap Tempo value. “Authtc” [Authentic] respects the natural pitch fluctuations inherent when changing a real delay pedal’s time knob; “Transpr” [Transparent] minimizes these artifacts.
2	1	Pedal 1 Jack	Determines whether the rear panel PEDAL/EXT AMP 1 jack acts as an additional expression pedal input (EXP 1) or as a dual amp control output for switching channels on external guitar amplifiers (EXT AMP 1 tip and/or sleeve).
	2	Pedal 2 Jack	Determines whether the rear panel PEDAL/EXT AMP 2 jack acts as an additional expression pedal input (EXP 2) or as a dual amp control output for switching channels on external guitar amplifiers (EXT AMP 2 tip and/or sleeve).
	3	Numbering	Determines whether presets appear as 32 banks of four (A B C D) or are numbered 000-127 (convenient when recalling presets via MIDI program change messages).
3	1	Snap Reselect	Determines the behavior when pressing a Snapshot mode footswitch again after loading its assigned snapshot. “Reload” (the default) simply reloads the stored state of the footswitch’s assigned snapshot again. “Toggle Previous” toggles between loading the previously selected snapshot and the footswitch’s assigned snapshot.

Global Settings > Switches

Page	Knob	Parameter	Description
1	1	Stomp Select	When set to "Touch," touching a Stomp mode switch selects its assigned item(s) but pressing doesn't. When set to "Press," pressing a Stomp mode switch selects its assigned item(s) but touching doesn't (helpful if you insist on playing barefoot). When set to "Both," either touching or pressing will select the assigned item.
	2	Stomp Mode	By default, Stomp footswitch mode displays 6 stomps. You may also choose "4 Switch", which replaces FS1 and FS4 with  and  switches.
	3	Preset Mode	When set to "Moment[ary]," the HX Effects device returns to Stomp mode as soon as a Preset (or Snapshot) switch is pressed. When set to "Latch[ing]," the HX Effects device stays in Preset (or Snapshot) mode and the user must press MODE to switch back to Stomp mode.
2	1	Snpsht Mode	When set to "Moment[ary]," the HX Effects device returns to Preset mode (or Stomp mode if Pg 1 > Knob 3 [Preset Mode] is set to "Moment[ary]") after selecting a snapshot. When set to "Latch[ing]," the HX Effects unit stays in Snapshot mode until pressing  +  to return to Preset mode (or MODE/EDIT/EXIT to return to Stomp mode).
	2	Up/Dn Switch	When set to "Preset" or "Snapsht," FS1 (BANK ) and FS4 (BANK ) change to PRESET  /  or SNAPSHOT  /  , where pressing either switch instantly selects the next/previous preset or snapshot, without a bank queue. This is useful if you've programmed a fixed set list for your show, and just want to increment through presets or snapshots. At any time, press and hold both FS1 () and FS4 () to cycle through BANK  /  , PRESET  /  , and SNAPSHOT  /  .
	3	Swap Up/Down	When Stomp Mode Switches (see the page 1, knob 2 item above) is set to "4 Switches," use the "On" setting here to reverse the direction of the FS1 and FS4 BANK, PRESET, or SNAPSHOT up/down actions. <i>SHORTCUT:</i> Touch (but don't press) the Up and Down Stomp Mode FS1 and FS4 switches together for one second, and then press the "OK" foot-switch.

Global Settings > Pedals

Page	Knob	Parameter	Description
1	1	EXP 1 Polrity	If your external expression pedal appears to work backwards—for example, a Volume pedal block is loudest with the heel all the way down—set its polarity to "Invert."
	2	EXP 2 Polrity	
	3	EXP 1 Position	Determines whether each expression pedal position is recalled per snapshot, per preset, or applied globally. If you want a Volume Pedal or Wah to maintain its position when switching presets, set this to "Global."
2	1	EXP 2 Position	

Global Settings > MIDI/Tempo

Page	Knob	Parameter	Description
1	1	MIDI Base Ch	Sets the system base MIDI channel that the HX Effects hardware uses to receive and send MIDI communication via MIDI and USB. Note that MIDI messages assigned from the Command Center page can be set to any MIDI channel.
	2	MIDI Thru	When on, MIDI OUT also acts as a MIDI THRU; that is, it passes through any MIDI messages received at the MIDI IN jack.
	3	Rx MIDI Clock	Determines whether the HX Effects device responds to incoming MIDI beat clock received at its MIDI IN port, via USB, or whichever it senses first ("Auto"). If you don't want your HX Effects device to respond to MIDI clock at all, set this to "Off."
2	1	Tx MIDI Clock	Determines whether the HX Effects device transmits MIDI beat clock from its MIDI OUT port, via USB, or both. If you don't want your HX Effects device to transmit MIDI clock at all, set this to "Off."
	2	Tempo	Choose whether the Tap Tempo value is stored and recalled with each snapshot, recalled with each preset, or is applied globally across all presets and snapshots.
	3	BPM	Depending on the Knob 2 (Tempo) setting, this value is saved per snapshot, per preset, or globally.
3	1	USB MIDI	When on, the HX Effects device receives and transmits MIDI data via USB in the same capacity as its MIDI jacks.
	2	MIDI PC Rx	Determines whether the HX Effects device responds to incoming MIDI program change (PC) messages from its MIDI IN port, via USB, or both MIDI and USB. If you don't want your HX Effects device to respond to PC messages at all, set this to "Off."
	3	MIDI PC Tx	Determines whether the HX Effects device automatically sends MIDI program change (PC) messages from its MIDI OUT port, via USB, or both MIDI and USB when selecting presets. This setting does not affect the sending of any MIDI commands manually assigned from the Command Center page.
4	1	Snapshot CC Send	Determines whether HX Effects automatically sends matching MIDI CC messages (CC69) when selecting Snapshots. This is handy for simultaneously recalling Snapshots on another MIDI-connected Helix or HX device. This option is "Off" by default.

Global Settings > Displays

Knob	Parameter	Description
1	LED Ring	Determines whether the Stomp mode footswitches' colored LED rings appear dim when bypassed ("Lo/Br") or off when bypassed ("Off/Br"). When playing in bright sunlight, you may want to set this to "Off/Br" to increase contrast.
2	Tap LED	If you'd prefer to not see the red TAP footswitch LED constantly flashing, you can turn it off.

MIDI

The HX Effects device receives and transmits MIDI messages via its 5-pin MIDI In & Out connectors, as well as via USB, providing functionality with just about any MIDI hardware or software gear you may want to use with it. USB MIDI operation is functional with Mac & Windows computers, as well as with iPad or iPhone mobile devices.* Also see ["Global Settings > MIDI/Tempo"](#) for MIDI device options.



***NOTE:** For USB MIDI operation on Windows computers, it is necessary to download and install the **Line 6 HX Effects device driver**, available from line6.com/software. There is no driver installation necessary for connecting to an Apple Mac computer or an iPad or iPhone mobile device.

For connectivity with iPad and iPhone mobile devices, an Apple Camera Connection Kit USB adapter is required (sold separately).

Preset Recall via MIDI

To recall a preset, send a program change (PC) message to your HX Effects device on MIDI channel 1, according to the table below.



NOTE: The HX Effects device responds to MIDI channel 1 by default, but this can be changed from ["Global Settings > MIDI/Tempo"](#)

Bank	Preset A	Preset B	Preset C	Preset D
01	PC: 000	PC: 001	PC: 002	PC: 003
02	PC: 004	PC: 005	PC: 006	PC: 007
03	PC: 008	PC: 009	PC: 010	PC: 011
04	PC: 012	PC: 013	PC: 014	PC: 015
05	PC: 016	PC: 017	PC: 018	PC: 019
06	PC: 020	PC: 021	PC: 022	PC: 023
07	PC: 024	PC: 025	PC: 026	PC: 027
08	PC: 028	PC: 029	PC: 030	PC: 031
09	PC: 032	PC: 033	PC: 034	PC: 035
10	PC: 036	PC: 037	PC: 038	PC: 039
11	PC: 040	PC: 041	PC: 042	PC: 043
12	PC: 044	PC: 045	PC: 046	PC: 047
13	PC: 048	PC: 049	PC: 050	PC: 051
14	PC: 052	PC: 053	PC: 054	PC: 055

Bank	Preset A	Preset B	Preset C	Preset D
15	PC: 056	PC: 057	PC: 058	PC: 059
16	PC: 060	PC: 061	PC: 062	PC: 063
17	PC: 064	PC: 065	PC: 066	PC: 067
18	PC: 068	PC: 069	PC: 070	PC: 071
19	PC: 072	PC: 073	PC: 074	PC: 075
20	PC: 076	PC: 077	PC: 078	PC: 079
21	PC: 080	PC: 081	PC: 082	PC: 083
22	PC: 084	PC: 085	PC: 086	PC: 087
23	PC: 088	PC: 089	PC: 090	PC: 091
24	PC: 092	PC: 093	PC: 094	PC: 095
25	PC: 096	PC: 097	PC: 098	PC: 099
26	PC: 100	PC: 101	PC: 102	PC: 103
27	PC: 104	PC: 105	PC: 106	PC: 107
28	PC: 108	PC: 109	PC: 110	PC: 111
29	PC: 112	PC: 113	PC: 114	PC: 115
30	PC: 116	PC: 117	PC: 118	PC: 119
31	PC: 120	PC: 121	PC: 122	PC: 123
32	PC: 124	PC: 125	PC: 126	PC: 127



NOTE: When navigating Presets from the HX Effects hardware (PRESETS encoder, preset footswitches, PRESET ▲ / PRESET ▼, etc.), the HX Effects device automatically transmits a MIDI Program Change message corresponding to the selected preset. If you'd prefer not to automatically transmit PC messages, set ["Global Settings > MIDI/Tempo"](#) > MIDI PC Tx to "Off."

Snapshot Recall via MIDI

To recall a snapshot, send the HX Effects hardware a CC69 message according to the table below.

Snapshot	MIDI CC#	Value
1	69	000
2	69	001
3	69	002
4	69	003
Next	69	008
Previous	69	009

Block Bypass via MIDI

1. Press  to open the Menu.
2. Press the SIGNAL FLOW switch.
3. Turn the Big Knob to select the block for which its bypass you want to assign to incoming MIDI.
4. Press  and turn Knob 3 (MIDI Bypass) to assign an incoming MIDI CC message to turn the block on and off.

Incoming CC values 0-63 turn the block off; values 64-127 turn the block on. Note that some MIDI CCs are reserved for global functions and cannot be selected.

Parameter Control via MIDI

1. From Stomp mode, press and hold the knob 1~3 for the parameter you wish to control.
The HX Effects device jumps to the Controller Assign page and displays your parameter at Knob 1 (Parameter).
2. Press the LEARN switch and then send a MIDI CC message from your foot controller, keyboard, etc.

The CC number appears above Knob 3 (CC#).

 **NOTE:** To manually select a MIDI CC, turn Knob 2 (Controller) to select "MIDI CC" and Knob 3 to the desired MIDI CC number.

3. If desired, press  and turn Knob 1 (Min Value) and Knob 2 (Max Value) to set the range you wish to control.



TIP: To reverse controller behavior, swap the min and max values.

4. Press  to return to the Home screen.

MIDI Clock Send and Receive

HX Effects is able to send and receive MIDI clock via its MIDI 5-pin in/out, as well as via USB. MIDI Clock options can be enabled and configured within the ["Global Settings > MIDI/Tempo"](#) options.

MIDI Clock Send: Transmit MIDI clock to synchronize the tempo of external pedals, rack gear, and software to the current HX Effects TAP Tempo rate.

MIDI Clock Receive: Synchronize the HX Effects time-based effects (such as delay and modulation) to incoming MIDI clock from external DAW software, drum machines, keyboard workstations, or other modelers. Note that the HX Effects TAP footswitch will flash blue (instead of the default red) to indicate the incoming tempo rate when receiving MIDI clock.

MIDI CC

The HX Effects hardware has reserved specific MIDI CC messages for global functions; these CCs cannot be used as controllers. If you attempt to learn a CC message reserved for global functions (see ["Controller Assign"](#)), the following dialog appears:



This MIDI CC is used
for global functions!

MIDI CC#	Value	Function
Pedal and Footswitch Assignments		
1	0-127	Emulates EXP 1 Pedal
2	0-127	Emulates EXP 2 Pedal
49	0-127	Emulates FS1 NOTE: These MIDI CC 49~54 messages control the respective switch's function while in Stomp, Preset, or Snapshot footswitch mode
50	0-127	Emulates FS2
51	0-127	Emulates FS3
52	0-127	Emulates FS4

MIDI CC#	Value	Function
53	0-127	Emulates FS5
54	0-127	Emulates FS6
Looper Controls		
60	0-63: Overdub; 64-127: Record	Looper Record/Overdub switch (FS4)
61	0-63: Stop; 64-127: Play	Looper Play/Stop switch (FS5)
62	64-127	Looper Play Once switch (FS6)
63	64-127	Looper Undo/Redo switch (FS1)
65	0-63: Forward; 64-127: Reverse	Looper Forward/Reverse switch (FS3)
66	0-63: Full; 64-127: Half	Looper Full/Half Speed switch (FS2)
67	0-63: Off; 64-127: On	Looper block on/off (if available); also enters/exits Looper footswitch mode
Additional Controls		
64	64-127	"Tap" in a Tap Tempo value
68	0-127	Tuner screen on/off
69	0-3 8 9	Snapshot 1-4 select Next snapshot select Previous snapshot select* TIP: Set the "Global Settings > MIDI/Tempo" -Snapshot CC option to "On" if you want HX Effects to automati- cally send the respective CC69 value to MIDI Out.
70	0-63: Bypass; 64-127: On	All Bypass
71	0-127	Engages the MODE switch
72	0-63: Previous; 64-127: Next	Loads the previous or next preset
73	n/a	Reserved for future use
74		
75	0-127	Parameter Knob 1 (global control)
76	0-127	Parameter knob 2 (global control)
77	0-127	Parameter knob 3 (global control)

MIDI CC#	Value	Function
78	n/a	Reserved for future use
79		
80		
81	0-63: Page Left; 64-127: Page Right	Triggers the device's Page Left and Page Right buttons
128	0-63: Off; 64-127: On	Talent Boost

*MIDI Snapshot changes on CC69 received during preset loads are buffered and executed once the preset load is finished. This means that you can send a MIDI CC69 Snapshot change immediately after a PC message to effectively load a preset with a different Snapshot than with which it was saved.

